AutoPasser

Final Report Document

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Executive Summary

Purpose

AutoPasser is a hockey puck automatic passer, intended to allow hockey players to practice receiving passes by themselves. This is a commonly used concept in other sports, like baseball, football, and tennis, and has been used before for hockey goalies, but a device to practice this skill is new to the market.

It is very hard to learn how to play hockey without paying for lessons, or learning at a young age. There are many skills that require another person to learn, and the sport has a very steep learning curve. AutoPasser will allow for any user to practice drills that normally require a second person. The device can feed the user a puck in the form of a pass similarly to a second person. This removes the restraint of requiring another person or lessons to practice.

The first AutoPasser prototype from Fall 2022 was able to store up to 7 pucks and shoot them at over 25 mph. This semester's team improved upon last semester's prototype. The main deliverables laid out from our client were to maintain firing speed while downsizing the assembly, designing a turret system to change the angle at which pucks are fired, and to fully enclose the assembly. This semester, we were able to increase the firing speed up to 56 mph, downsize and enclose the entire assembly with a matte black acrylic chassis, and implement a turret system that would rotate the firing assembly up to 180 degrees. The team also added a remote that could start the firing process from afar.

This system is interdisciplinary between mechanical engineering, electrical engineering, and design. Members of the team produced high quality prototypes of their system in their respective disciplines of strength, and were also exposed and contributed to other disciplines.

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Background

One of the existing competitors is the Pro Paza Puck Passing Machine. This machine can load up to 20 pucks and rotate 360 degrees, however it is $1560 and several feet tall, with much less customizability than desired from our client. The market for a smaller, cheaper, and more customizable hockey puck passing machine still exists.

The background research for this semester carried on from last semester's findings. Last semester's team based the firing system off existing mechanisms in popular baseball and football passing machines. Two flywheels would spin in opposite directions and a ball would be loaded in and fired at high velocity. We decided to apply the same mechanism to this semester's prototype. Other research included looking into the flywheels themselves and seeing how size, hardness, and friction affected the output speed of a puck. The team downsized the wheels and tried three different hardnesses based on this research. The team conducted experiments to determine which hardness was optimal.

The requirements and constraints for this product were determined by our client with input from the team.

1. This product shall hold at least 7 hockey pucks
2. This product shall deliver pucks at controlled intervals to a firing subsystem
3. This product shall fire pucks at a speed of at least 20 mph
4. This product shall decrease in footprint by at least 1" in each dimension while maintaining the same or improved output
5. This product shall be able to rotate at least 90 degrees
6. This product shall be stationary, and not affected by momentum when firing
7. This product shall have a method for users to control speed, firing intervals, and angle
8. This product shall be durable and long lasting
9. This product shall be easy to move
10. This product shall be easy to load pucks into
11. This product shall be intuitive to use
12. This product shall have well routed and organized wiring

The priority deliverables and reach deliverables were also determined by the client with input from the team.

Priority Deliverables

1. Turret Subsystem

Reach Deliverables

1. ~~Full Battery Power~~

2. Reduced footprint with custom PCBA while maintaining puck speed
3. Fully enclosed assembly

2. Remote Control Capability
3. Redesigned Puck Loading System

The battery-powered reach deliverable was scrapped within a few weeks because it would add too much size and weight.
Last semester's prototype had a loading system that could hold up to 7 pucks. This semester, with the full loading system in place, the prototype can hold 9 pucks. The reach deliverable was originally intended to be an overhaul to hold 10+ pucks, but this would have been contradictory to the priority deliverable of reducing the footprint, so 9 pucks was the goal instead.
This semester, we were able to achieve all three priority deliverables and two out of three of the reach deliverables laid out at the beginning of the semester.

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Systems Overview

- AutoPasser is a machine created to allow a user to practice receiving passes by themselves. This machine can hold up to 9 pucks and rotate the angle at which pucks are fired.
- The user will start by stacking pucks on top of each other, then pressing the stack into a loading column. The loading column will twist into the column mounted to the machine, which will then load 7+ pucks. The user will change settings for firing speed, angle, interval, and delay with buttons by the screen. The user will then stand far away and press the button on the remote to start the firing process. Based on the settings the user chose, the machine will change the angle at which pucks are fired until the machine runs out of pucks.

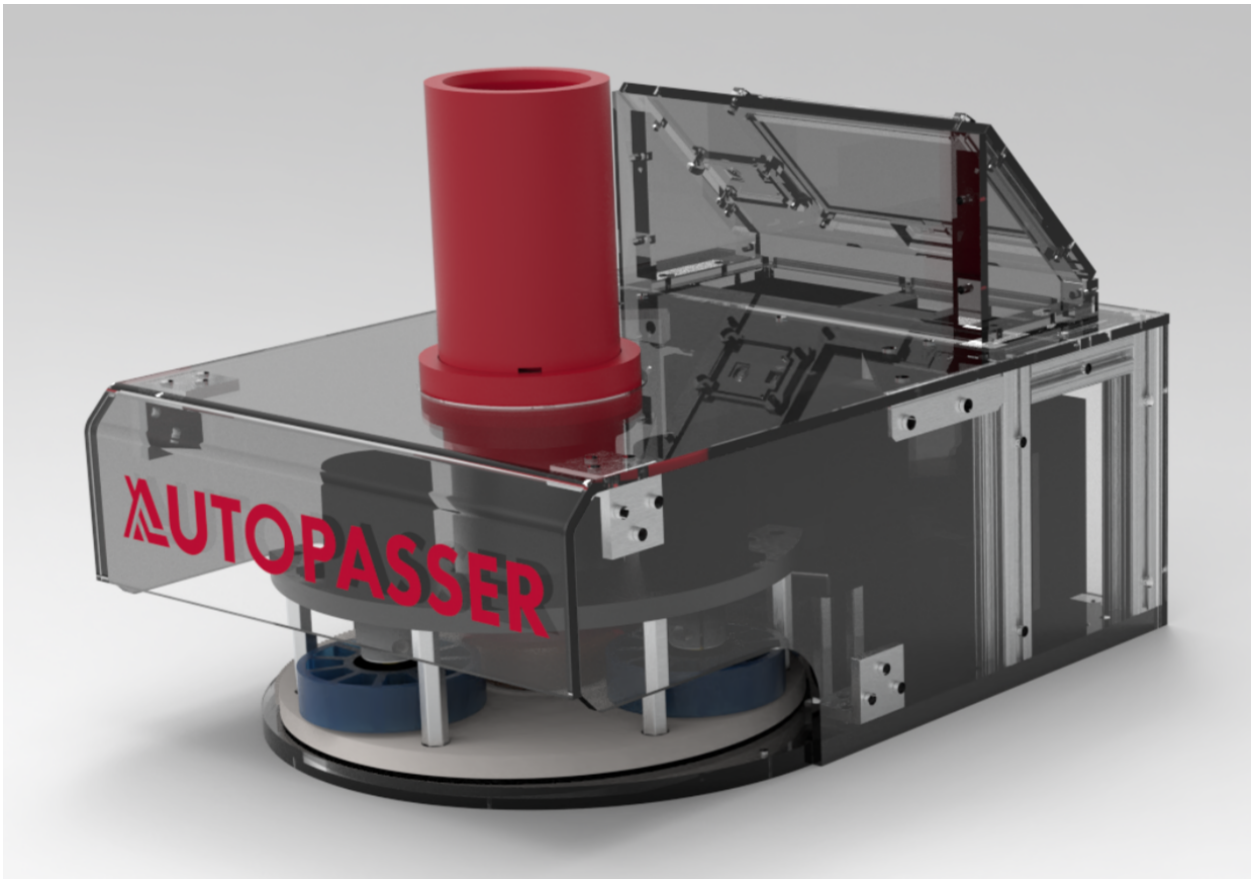

Figure 1: AutoPasser Full Assembly Render

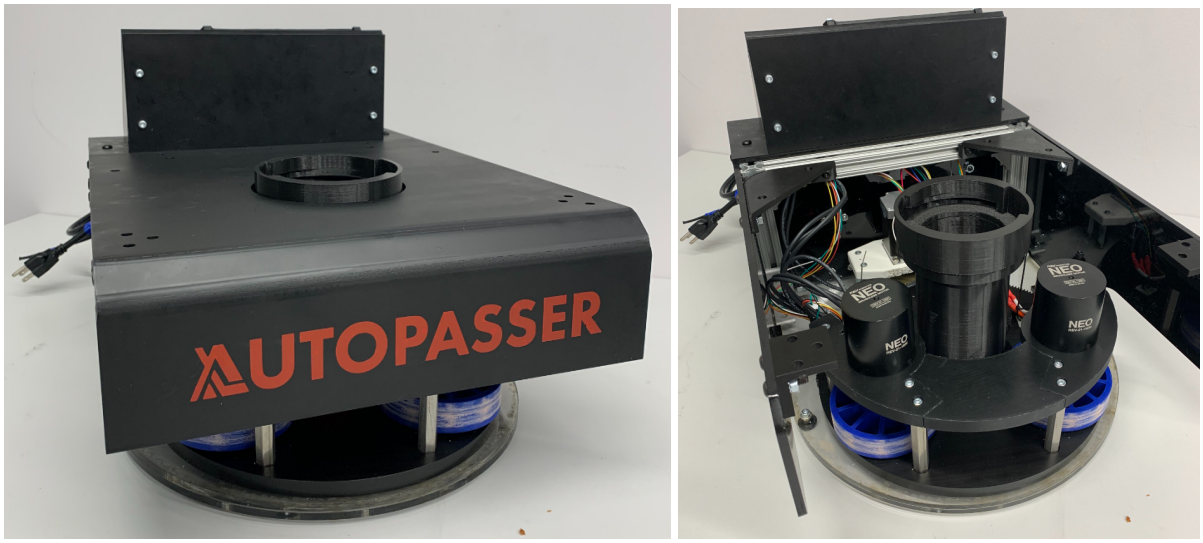

Figure 2: Final Prototype Full Assembly vs Deconstructed Internal View

- AutoPasser is an interdisciplinary machine that combines mechanical engineering, electrical engineering, and industrial design.
  - The baseplate and 80/20 frame help to mount electrical components. The baseplate holds a lazy susan that a delrin gear is mounted to. This delrin gear holds the loading and firing system. The delrin gear is rotated via a driving gear controlled by a stepper motor. Once rotated, the firing process starts. A linear actuator mounted to the delrin gear pushes pucks towards the flywheels. Each flywheel is controlled by a NEO motor.
  - The user can change settings for firing speed, interval, delay, and angle using the buttons by the screen. This will change how the NEO motors, linear actuator, and stepper motor run.
  - To fully enclose the assembly, a chassis made of laser cut acrylic that was spray painted matt black is mounted to the 80/20 and baseplate.

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Mechanical Design

Assembly Overview

The top level assembly contains several subsystems with electronic and mechanical integration to achieve an automated hockey puck launching system. From the top down, the system includes a custom chassis to house and protect all internal components, a redesigned external facing electronics mounting panel for the interactive screen, buttons and switches, signaling LED's, and an emergency stop. Inside, there is an additional electronics mounting box in the back of the system, and the main turret system which contains nearly all of the mechanical components that allow pucks to be loaded and launched with varying degrees of speed, angle, etc. Main components are called out in the figures below.

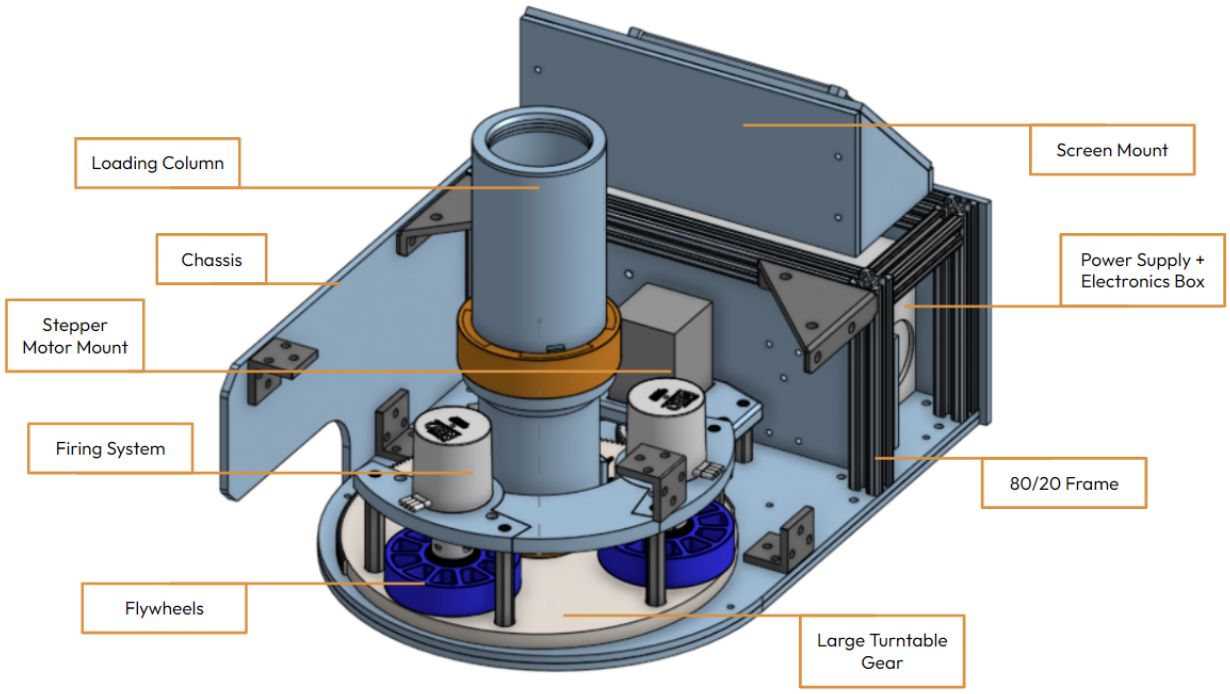

Figure 2: Labeled Top Level Assembly

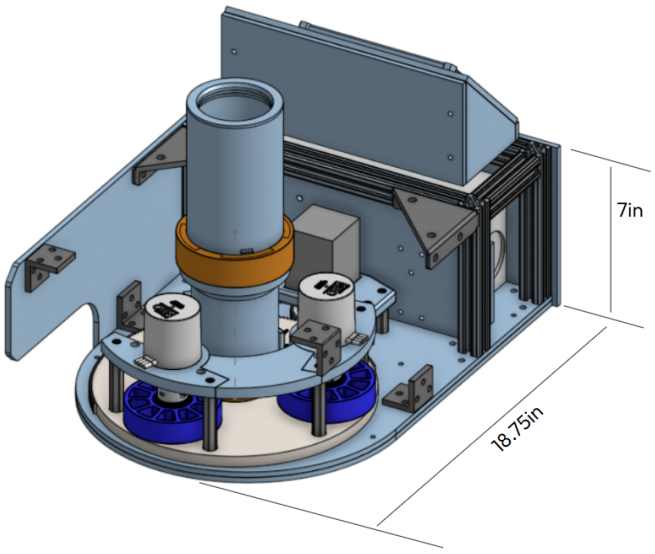

Figure 2: Major Dimensions of Top Level Assembly

This design was developed after several iterations and evaluations of capabilities and limitations, but succeeded due to its reduction in form factor from the existing system, its ease of manufacturing and assembly, having the smallest gap between the ground and height of puck launch, and it requiring fewer moving components than other initially proposed designs.

As mentioned in the Systems Overview, the main objective of the client was to develop a system that automatically launches hockey pucks and adjusts the angle and speed at which they are launched by dialing some knobs. The core deliverables for the engineering team included development of remote control capability, creating a turret system that would allow the angle of the puck launch to be adjusted, to reduce the overall footprint of the design, and to fully enclose all electronic and otherwise exposed or moving components with a custom chassis. Core deliverables for the design elements of the project included doing a rebrand for AutoPasser with new logos, wordmarks, typefaces and a color scheme, as well as wireframing and designing the GUI for a smoother user experience with the system's screen. Every component included in the system is referenced and listed in the Bill of Materials (BOM) in the Appendix that can be found at the end of the document.

While the mechanical design and overall assembly were ultimately successful, there were some resulting shortcomings of the design to be adjusted in a further iteration. Each of these are addressed in more detail with next steps under their respective subassembly sections. In summary, these shortcomings include manufacturability of the chassis and external electronics mount, the loading column causing unwanted rotation of the turret when being inserted, and some mechanical adjustments to assist in electronics controls to prevent failure.

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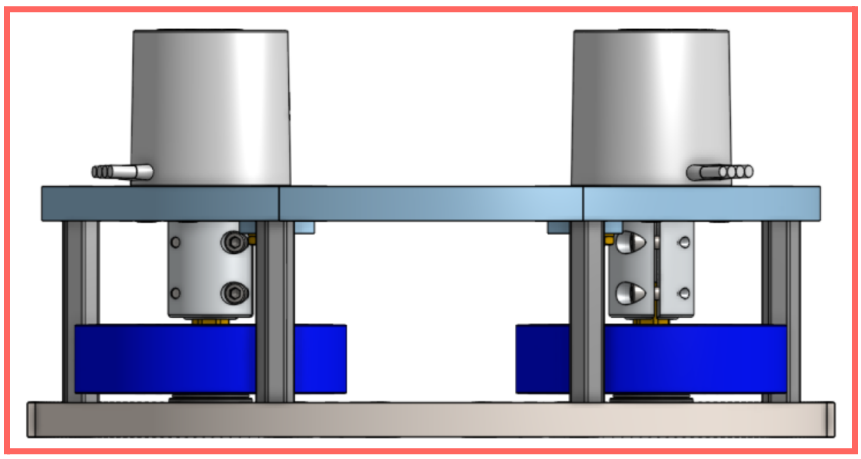
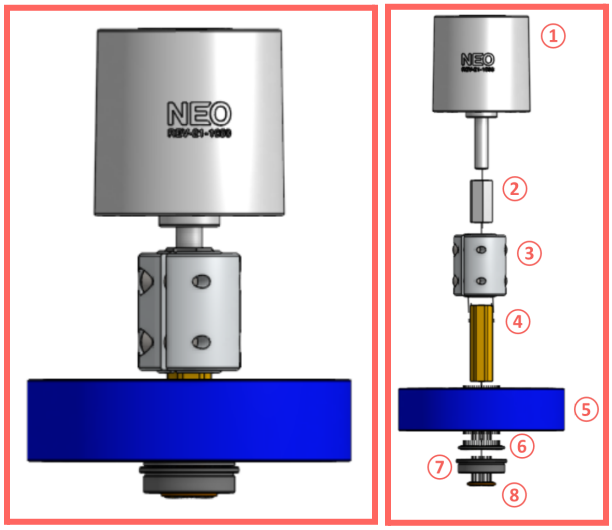
Firing System



Flywheel Subassembly (2x)



1. [NEO Brushless Motor V1.1] - *12-volt DC motor that drives the system's rotation*
    - Nominal Voltage: 12 V
    - Empirical Free Speed: 5,676 RPM
    - Empirical Stall Current: 105 A
    - Empirical Stall Torque: 2.6 Nm
    - Typical Output Power at at 40 A: 380 W
    - Output Shaft Diameter: 8mm (2mm keyed)
    - Body Length: 58.25 mm (2.3")
    - Weight: 0.938 lbs (0.425 kg)
    - Mounting Holes: #10-32 tapped
2. [8mm to 1/2" Hex Shaft Adapter] - *Converts shaft output from 8mm round to 1/2" hex*

- Bore: 8 mm Round
- Length: 1 in.
- Material: Steel
- Weight: 0.03 lbs
- Width: 0.497 - 0.499 in.

3. **1/2" Hex and Keyed Shaft Coupling** - *Couples the adapter and the hex shaft*
   - Bore: 0.5 in. Hex
   - Length: 1.5 in.
   - Material: 6061-T6 Aluminum
   - Outside Diameter: 1.25 in.
   - Weight: 0.139 lbs

4. **1/2" Rounded Hex Shaft - 2.0" - Tapped** - *Translates rotation to the flywheels*
   - Length: 2in
   - Hex Size: 0.5in
   - Thread Size (both ends): 10-32
   - Thread Depth (both ends): 0.5in
   - Material: Aluminum 7075 (Black Anodized Finish)

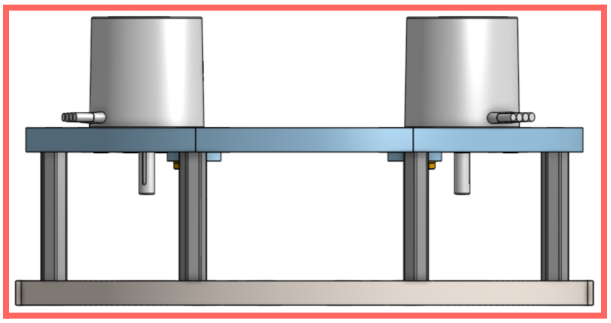
5. **4" Compliant Wheel, 1/2" Hex Bore, 50A Durometer** - *Translates rotational motion to of system to linear motion of puck*
   - Bore: 1/2 in. Hex
   - Diameter: 4 in.
   - Durometer: 50A
   - Maximum Rotation: 9,000 RPM
   - Weight: 0.266 lbs
   - Width: 1 in.

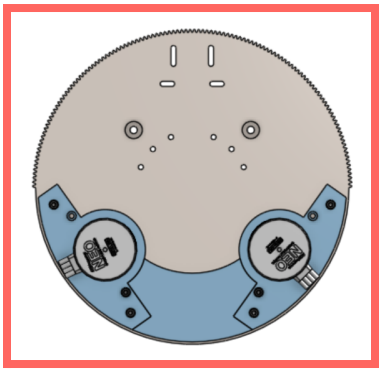
6. **Needle-Roller Thrust Bearing for 5/8" Shaft Diameter, 1-1/8" OD** - *Reduces friction between flywheel hub and flanged bearing, allowing for rotation*
   - For Shaft Diameter: 5/8"
   - ID: 0.627"
   - OD: 1 1/8"
   - Thickness: 5/64"
   - Material: Steel
   - Maximum Speed: 11,000 rpm

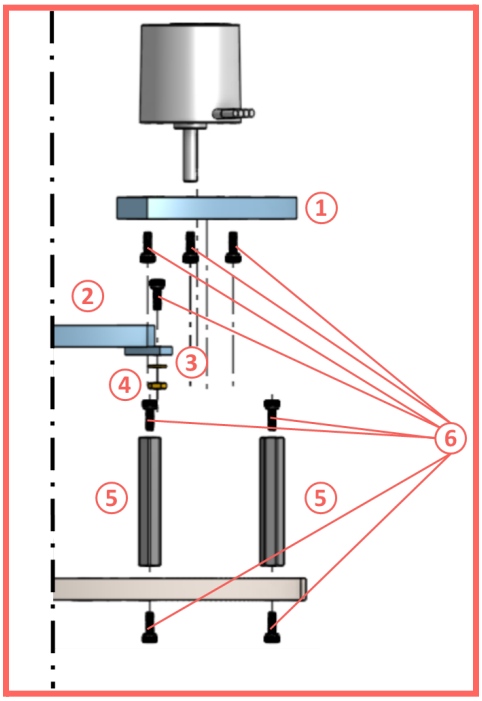
7. [1/2" Hex ID 1.125" OD Shielded Flanged Bearing] - *Constrains shaft movement in the X and Z directions while allowing for rotation in the Z-plane*
    - Bore: 0.5 in. Hex
    - Flange Thickness: 0.062 in.
    - Inside Diameter: 0.5 in.
    - Material: Steel
    - Max Speed: 32,000 RPM
    - Outside Diameter: 1.125 in.
    - Overall Thickness: 0.312 in.
    - Weight: 0.046 lbs
    - Width: 4 mm

8. [Needle-Roller Thrust Bearing for 1/4" Shaft Diameter, 11/16" OD] - *Reduces friction between shaft and base plate, allowing for rotation*
    - For Shaft Diameter: 1/4"
    - ID: 0.252"
    - OD: 1 1/16"
    - Thickness: 5/64"
    - Material: Steel
    - Maximum Speed: 20,000 rpm

Mounting Subsystem

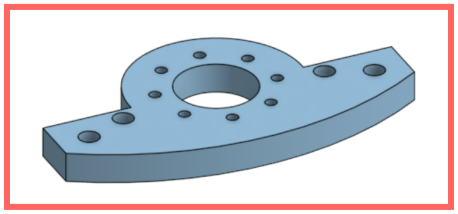


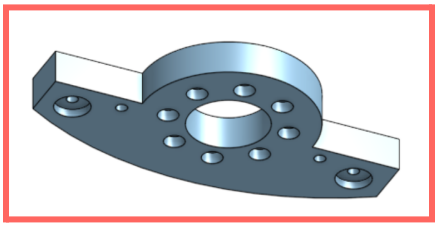




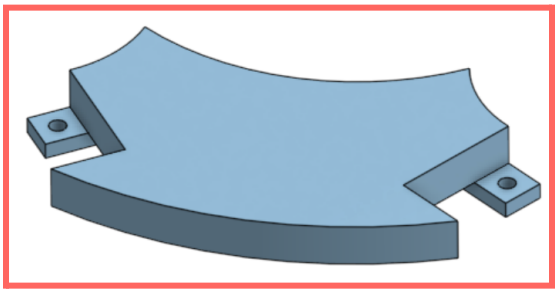
*See the Flywheel Subassembly for motor specifications (top) and the Turret System for base plate specifications (second from bottom). These are included in this depiction for visual continuity and are not part of the actual assembly.

1. **Motor Mounting Plate (2x)** - *Fixes motor in place*
   - Material: PLA (50% Infill)
   - Maximum Thickness: 0.5 in.
   - Maximum Length: 6.0615 in. (flat edge)
   - Maximum Width: 4.6902 in. (outside arc to inside arc)
   - Body Diameter: 11.5 in.
   - Through-Hole Size: 10-32 (Close Fit)
   - Counterbore Size: ⌀ 0.35” (varying depths)
   - Motor Clearance Hole: 35mm





2. **Bridge Plate** - *Connects motor mounting plates to limit undesired movement in the Z-plane*
    - Material: PLA (50% Infill)
    - Maximum Thickness: 0.6875 in. (0.5 in. plate; 0.1875 in. mounting legs)
    - Maximum Length: 5.6512 in. (between mounting legs)
    - Maximum Width: 4.8549 in. (outside arc to inside arc)
    - Body ID: 6.3655 in.
    - Body OD: 11.5 in.
    - Through-Hole Size: 10-32 (Close Fit)



3. 18-8 Stainless Steel Washer for Number 10 Screw Size, 0.203" ID, 0.438" OD **(2x)**- *Protects bridge plate from indentation from the nut*
    - Material: 18-8 Stainless Steel
    - For Screw Size: No. 10
    - ID: 0.203"
    - OD: 0.438"
    - Thickness: 0.025"-0.040"
    - Washer Type: Flat

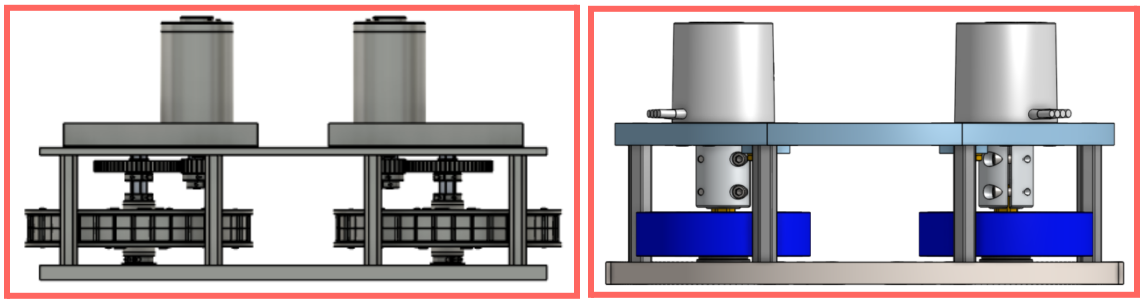
4. Low-Strength Steel Hex Nut, Zinc-Plated, 10-32 Thread Size **(2x)** - *Constrains screws holding bridge plate in place*
    - Material: Zinc-Plated Steel
    - Thread Size: 10-32
    - Thread Type: UNF
    - Width: 3/8"
    - Height:1/8"
    - Nut Type: Hex

5. Female Threaded Hex Standoff, 18-8 Stainless Steel, 1/2" Hex, 3" Long, 10-32 Thread **(4x)** - *Elevates motor mount to appropriate height for assembly*
    - Shape: Hex
    - Material: 18-8 Stainless Steel
    - Hex Size: 1/2"
    - Length: 3"
    - Thread Size: 10-32
    - Threading: Partially Threaded

- Min. Thread Length: 1/2"
- Gender: Female
- Tensile Strength: 80,000 psi

6. Black-Oxide Alloy Steel Socket Head Screw, 10-32 Thread Size, 1/2" Long **(18x)** - *Fixes various components of assembly in place*
    - Head Type: Socket
    - Drive Style: Hex
    - Thread Size: 10-32
    - Thread Type: UNRF
    - Length: 1/2"
    - Threading: Fully Threaded
    - Thread Spacing: Fine
    - Head Diameter: 5/16"
    - Head Height: 0.19"
    - Drive Size: 5/32"
    - Material: Black-Oxide Alloy Steel
    - Tensile Strength: 170,000 psi

Evolution of Firing System



The previous semester's firing assembly (left) versus the current semester's firing assembly (right).

The main goal for the firing system was to **decrease footprint size and assembly complexity** whilst **maintaining or increasing firing velocity** from 20 miles per hour. The first major alteration from the previous design was the reduction of flywheel size from 8 inches to 4 inches. This facilitated a 42% decrease in major dimension length from around 19 inches to only 11 inches. The wheel material was also drastically changed from brittle plastic covered with a griptape tread to a compliant polymer with a much lower durometer of 50A. This allowed for increased compression of the wheel, which facilitated a more efficient transfer of energy from the flywheels to the puck during firing.

Another major change to the system was the drive system. To reduce mechanical and assembly complexity, the drivetrain from the previous design was replaced in favor of a direct-drive system. The large CIM motors were also replaced with smaller, more efficient brushless motors. In conjunction with the lower durometer flywheels, these motors facilitated a 100% increase in maximum firing velocity from 28 miles per hour to 56 miles per hour. The overall height of the drive system was reduced by an inch as a result of the elimination of the drivetrain in conjunction with the implementation of a thrust bearing in place of a collar on the bottom of the flywheels.

The final major change between the two firing systems were the mounting features. The previous semester's single top plate was divided into three different plates–one for each motor and an additional plate to connect the two motor mount plates together. This made it significantly easier to align the motor shafts with the adapters, which helped cut assembly time from hours to minutes.

Manufacturing Recommendations

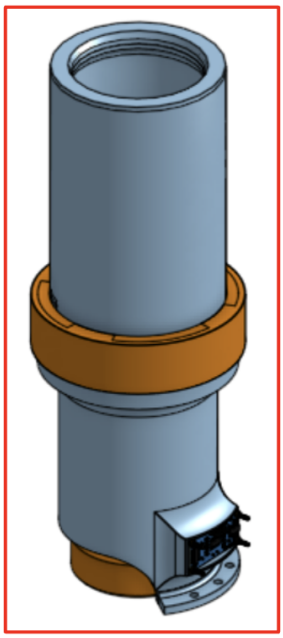
A major focus of the design of the firing subassembly was the emphasis on ease of manufacturing and assembly. As a result, the majority of the parts were purchased off-the-shelf as opposed to being manufactured. The major exceptions to this are the motor mount plates and the bridge plate. Because these plates were designed to be 3D printed using PLA, it is likely that their thicknesses could be reduced if they were to be manufactured with a more robust material such as ABS or POM.

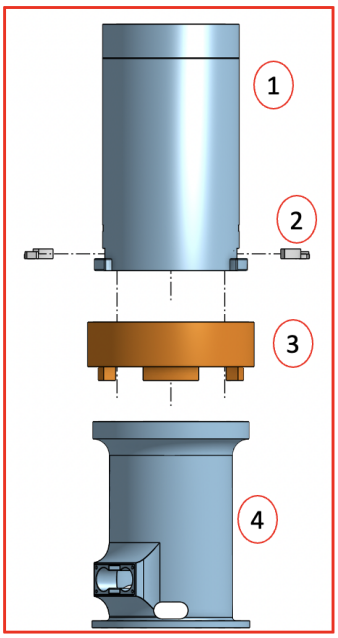
In lower volumes, the motor mount plates could be machined using subtractive manufacturing processes. For example, the exterior geometry could be achieved by laser cutting or water jetting from a large, half-inch thick ABS or POM sheet before features requiring bores would be added in a CNC mill. For high-volume production, injection molding would likely become the preferred manufacturing method.

Due to its unusual geometry, the bridge plate could not be produced from a single sheet of material like the motor mounting plates. This would pose an issue for low-volume manufacturing, and would likely require a design change to decrease manufacturing time and cost if production using a 3D printer was deemed undesirable. A possible design change would be the removal of the mounting legs, replacing them with off-the-shelf metal or plastic brackets that could be affixed to the plate. This would allow for the bridge plate to be manufactured via the same method as the motor mounting plates–using a subtractive method to cut out the geometry from a large plastic sheet before adding further features with a CNC mill. With this in mind, the bridge plate would benefit most from injection molding, as the utilization of injection molding would no longer facilitate the need for a change to the design of the part. Therefore, high-volume production would undoubtedly utilize injection molding.

Loading System

The loading system was reworked from last semester to decrease the radial size on the rotating platform. To do this, the torsional mechanism that was once on the bottom of the loading system was moved to the top of the loading column. The pucker upper was also redesigned to decrease footprint and remove the need for rubber tubing. The loading system can handle a total of 10 pucks. If more than 10 pucks were loaded in, the linear actuator would not have enough power to push the bottom puck into the flywheels. Due to time and manufacturing constraints, some of the final tolerances are not fully optimized.

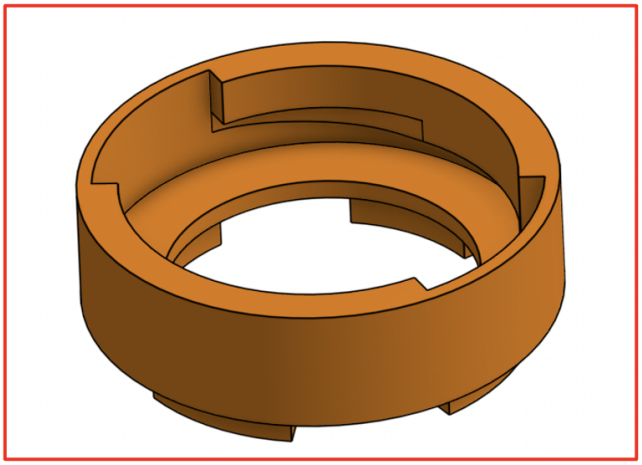




1. Pucker Upper - Plastic Column with a compliant lip at the top that allows pucks to be pushed through. This part needs to be printed using resin 3D printing for durability reasons. The tolerancing was also done using Formlabs Grey Pro Resin. Due to manufacturing constraints, this part was split into two. The smaller compliant part was printed in Grey Pro while the larger main column was printed in PLA.

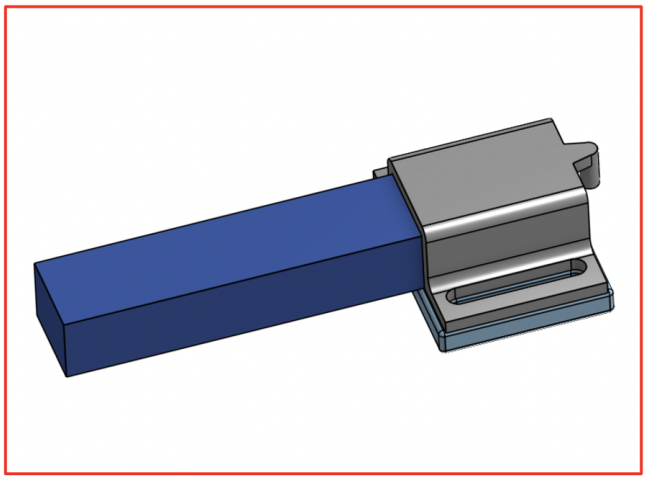
2. Puck Stoppers - Plastic pieces that attach to a torsional spring. The other part of the spring is attached to the Pucker Upper. When the springs are at rest, the puck stoppers are rotated into the middle of the column stopping the pucks from falling through.

3. Torsional Gate - Plastic component that the Pucker Upper twisted into. When the user twists the column into the gate, the Puck Stoppers rotate, allowing the pucks to fall through.



4. Loading Column - Plastic column that is fixed to the turret gear. This column has enough vertical space to fit 6 pucks and also has an ultrasonic sensor housing to let the user know if there are pucks in the loading system. The loading column is bolted into the turret gear using 10-32 screws. The ultrasonic sensor is glued into the column.

Once the pucks were at the bottom of the loading column, a linear actuator is used to push the puck into the firing assembly at the correct time. The picture below shows the actuator mount holding dummy cad of the linear actuator.



Alternative Designs

Throughout the semester, many different design ideas were thought up for the loading system. One main goal was to decrease the size of last semester's linear actuator. One way to do this was to use a DC motor and a mechanical linkage like a piston in an engine. We chose to go away from this idea once we found the smaller linear actuator on Amazon.

There was a lot of time spent on trying to rework the puck columns so the user could load in 20-30 pucks. Different conveyor belt designs were drawn up but ultimately we decided that it was out of scope for the project and would add too much complexity to both the mechanical and electrical engineers. However, the current design can be easily altered to take on a larger loading system. To do this the user would only need to keep the puck column that is attached to the base plate and can design a system that feeds into this column. Since the column is at the center of rotation, this larger loading system would not need to move with the plate.

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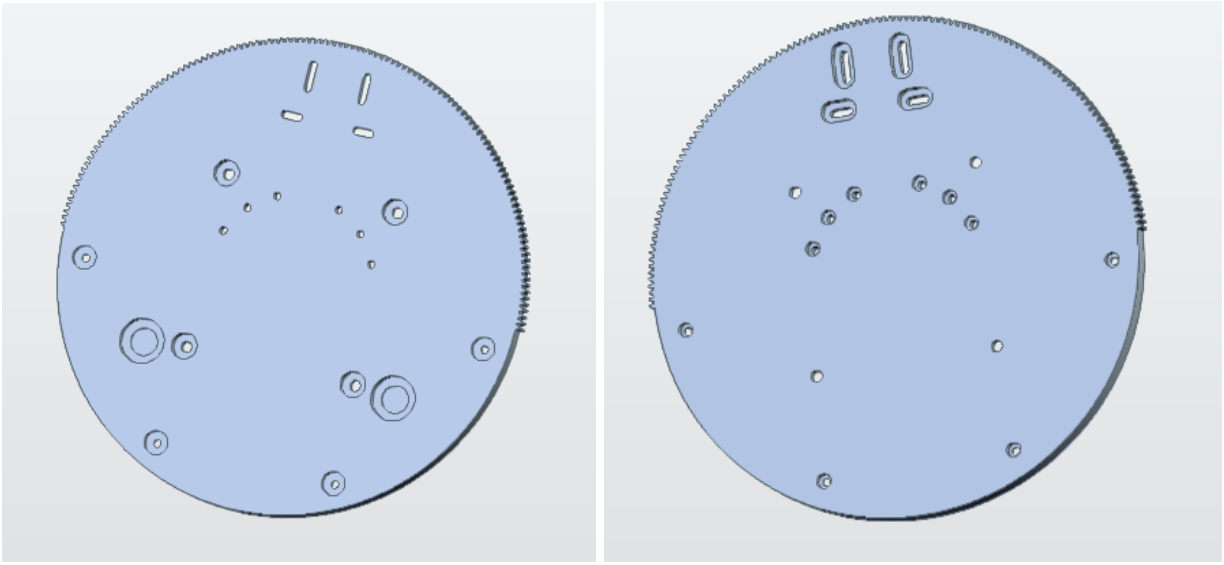
Turret System

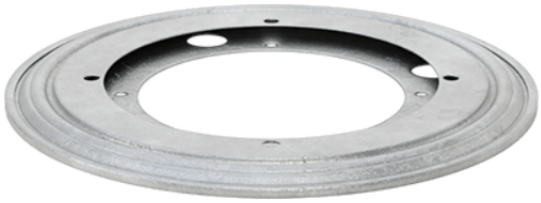
The turret system contains the large mounting gear with all firing and loading features attached, as well as a stepper motor mount that mates to a smaller driving gear and a linear actuator with custom mounts. All major components are detailed below, as well as labeled images.

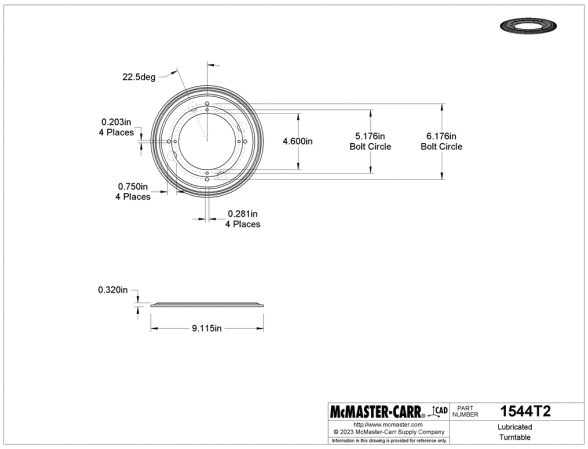
1. **Large Mounting Gear:**
    - Attached to base plate, houses all firing and loading features while adding rotational functionality to the system
    - Material: Black Delrin 150
    - Thickness: 0.5in
    - Diametral Pitch/Pressure Angle: 20/14.5
    - Pitch Diameter: 11.8in.
    - Number of Teeth: 236 for total gear, actual gear is toothed halfway (118 teeth)
    - Cost: $292.07



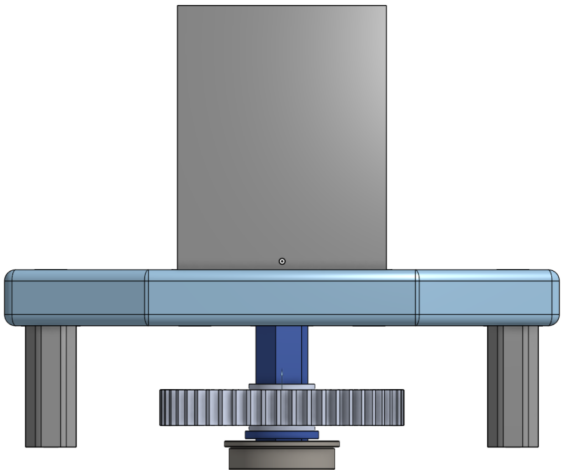
Top View (Left), Bottom View (Right)

2. **Turntable**
    - Couples Mounting Gear to Baseplate, allowing rotational motion in the plane parallel to the flat gear face.
    - McMaster 1544T2





3. **Stepper Motor Mount:**
    - Attached to base plate, securely mounts the stepper motor and provides a D-Shaft to Hex Adapter to interface a direct drive to the smaller pinion gear
        i. Modified dimensions to reduce overall form factor of the base plate as much as possible - hex standoffs reduced to 1.5”
    - Material: FormLabs Resin Tough 2000
    - Thickness: 0.5in
    - Width: 11.8in
    - Depth: 3in
    - Cost: $292.07



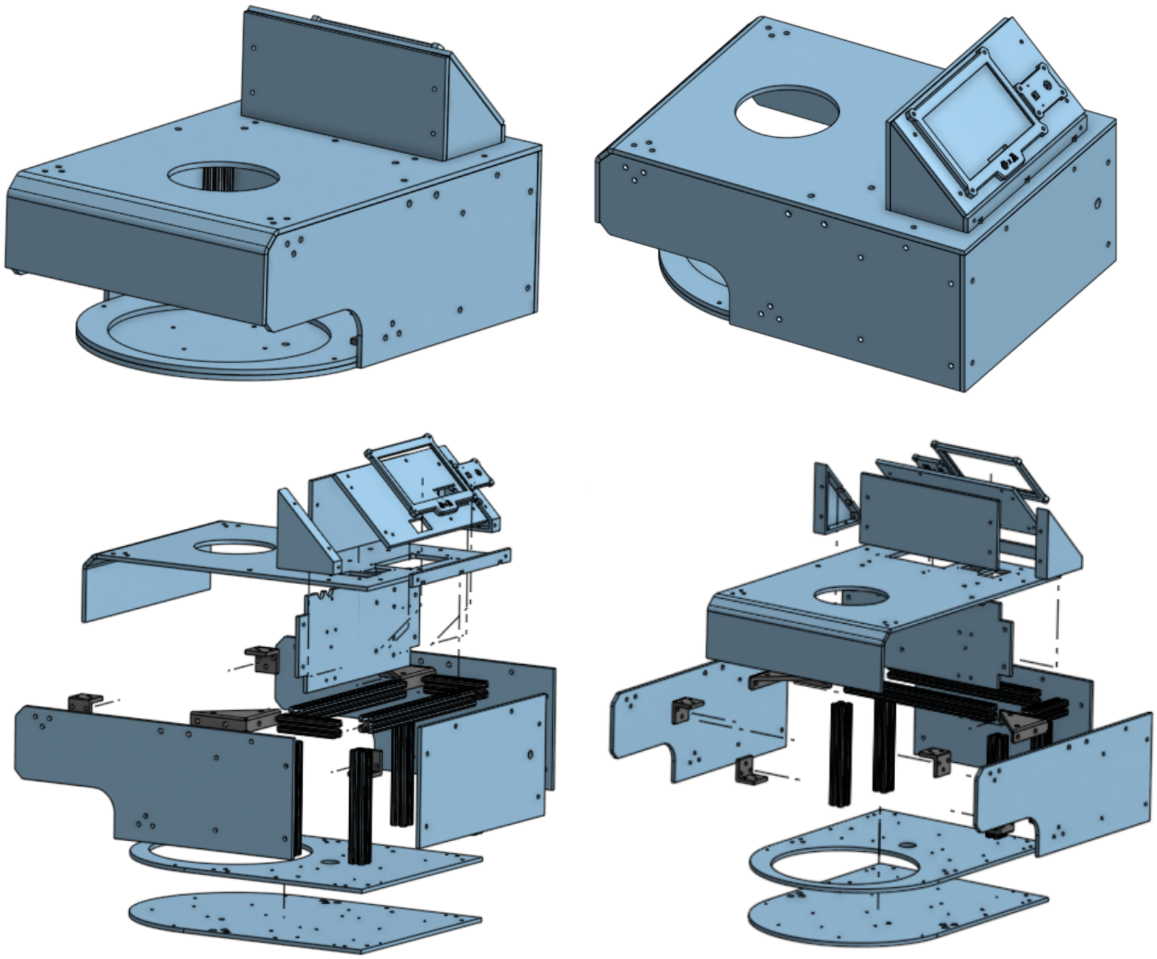
4. **Pinion Gear:**
    - Interfaces with the base plate and the d-shaft to hex-shaft adapter from the stepper motor and mount, provides a 4.72:1 gear ratio
    - Material: 7075-T6 Aluminum
    - Thickness: 0.5in
    - Diametral Pitch/Pressure Angle: 20/14.5
    - Number of Teeth: 50
    - Cost: $16.99

Alternative Designs

Alternative designs were explored for the turret system in several areas, including looking into larger gear trains, belt and pulley systems, and chain and sprocket systems. While any of these options would have been able to achieve the rotary motion required, there were certain features that would cause loss of important functionality or they were beyond the scope of the project in terms of budget and timeline.

Regarding the option to use a belt and pulley system in particular, those were found during research to be better suited for rotary systems that require continuous movement rather than frequent stopping and starting, which is what the turret system design would require. In addition, there was the potential for slip if the coefficient of friction of the belt could not be matched to the system requirements in time. A chain and sprocket system was briefly considered by the team, but ultimately not pursued due to limitations with budget and manufacturing - an additional top plate similar to that of the large mounting gear would still have to be designed and manufactured to contain all of the loading and firing components. Some larger gear trains were also considered but they ultimately raised the cost of the overall assembly and required more space when the desired goals could be achieved with a simple gear and pinion system using direct drive from the stepper motor to the pinion.

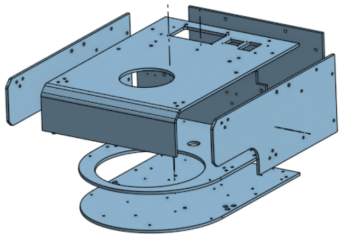
Chassis Design



One of our main objectives as outlined by the client was to create housing to enclose all the hardware and electrical components for both safety of the user and aesthetics. After several brainstorming sessions and iterations, we developed a robust chassis that encompassed all components while providing an optimal user experience. Our chassis design consists of three main assemblies: the exterior plates, the LCD display mount, and the chassis framing.

generate

Exterior Plates



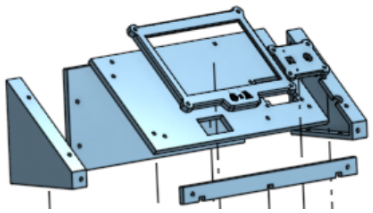
1. **Bottom Exterior Plate** - *Encloses the bottom of the chassis*
    - Material: Clear ¼” Acrylic
    - Length: 19.75”
    - Width: 14”
    - OD: 14”
    - Screw Clearance: 0.438”, 0.257”, 0.2”
2. **Bottom Interior Plate** - *Provides support for interior-mounted components*
    - Material: Clear ¼” Acrylic
    - Length: 19.75”
    - Width: 14”
    - OD: 14”
    - ID: 10”
    - Screw Clearance: 0.27”, 0.257”, 0.2”
3. **Side Plates (2x)** - *Enclose the side of the chassis*
    - Material: Black ¼” Acrylic
    - Finish: Flat Black Spray Paint
    - Height: 7.5”
    - Width: 19.93”
    - Screw Clearance: 0.263”
4. **Top Plate** - *Encloses the top and front of the chassis*
    - Material: Black ¼” Acrylic
    - Finish: Flat Black Spray Paint
    - Length*: 24”
    - Width: 14.5”
    - Screw Clearance: 0.263”

    *measured in sheet metal flat view
5. **Back Plate** - *Encloses the back of the chassis*
    - Material: Black ¼” Acrylic
    - Finish: Flat Black Spray Paint
    - Length: 7.5”
    - Width: 14.5”

- Screw Clearance: 0.263”

6. **Electronics Mounting Plate** - *Provides face for mounting electronic devices*
    - Material: Black ¼” Acrylic
    - Length: 7.5”
    - Width: 14.5”
    - Screw Clearance: 0.26”, 0.1”, 0.197”

LCD Display Mount

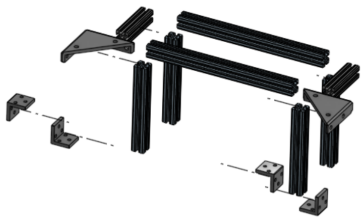


1. **Side Plates (2x)** - *Enclose the sides of the LCD mount*
    - Material: Black PLA
    - Height: 4.82”
    - Width: 4.04”
    - Viewing Angle: 45°
    - Screw Clearance: 0.268”
2. **Front Plate** - *Encloses the front of the LCD mount*
    - Material: Black ¼” Acrylic
    - Finish: Flat Black Spray Paint
    - Length: 0.79”
    - Width: 11.38”
    - Screw Clearance: 0.268”
3. **Top Plate** - *Provides support for LCD display*
    - Material: Black ¼” Acrylic
    - Finish: Flat Black Spray Paint
    - Length: 5.71”
    - Width: 11.38”
    - Screw Clearance: 0.268”
4. **Back Plate** - *Encloses the back of the LCD mount*
    - Material: Black ¼” Acrylic
    - Finish: Flat Black Spray Paint
    - Length: 5.71”
    - Width: 4.82”
    - Screw Clearance: 0.268”
5. **LCD Frame** - *Provides an exterior frame around the LCD display*

- Material: Black ¼" Acrylic
- Finish: Flat Black Spray Paint
- Length: 5"
- Width: 7.09"
- Cutout Length: 3.95"
- Cutout Width: 6.33"
- Screw Clearance: 0.268"

6. **Button Mount** - *Provides an exterior frame around the buttons*
   - Material: Black ¼" Acrylic
   - Finish: Flat Black Spray Paint
   - Length: 1.77"
   - Width: 2.56"
   - Screw Clearance: 0.268", 0.1"

Chassis Framing



1. **Vertical 80/20 Supports (4x)** - *Provide support in the Y-direction of electronics box*
   - Length: 7"
   - Framing Type: T-Slot
   - T-Slot Framing Component: Rail
   - Rail Profile: Single
   - Single Rail Profile Style: Four Slot
   - Rail Height: 1"
   - Rail Width: 1"
   - Rail Construction: Solid
   - T-Slot Width: 0.256"
   - Center Hole Diameter : 0.205"
   - Material: 6105 Aluminum

2. **Horizontal 80/20 Supports (2x)** - *Provide support in the X-direction of electronics box*
   - Length: 12"
   - Framing Type: T-Slot
   - T-Slot Framing Component: Rail

- Rail Profile: Single
- Single Rail Profile Style: Four Slot
- Rail Height: 1"
- Rail Width: 1"
- Rail Construction: Solid
- T-Slot Width: 0.256"
- Center Hole Diameter : 0.205"
- Material: 6105 Aluminum

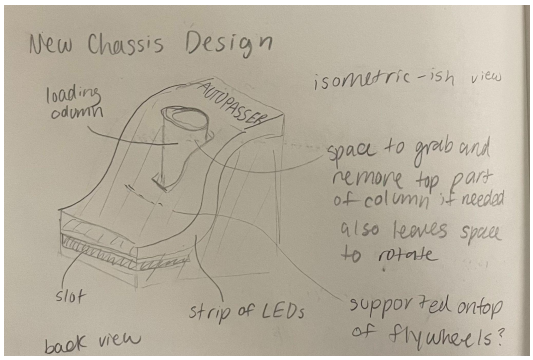
3. **Connecting 80/20 Supports (2x)** - *Provide support in the Z-direction of electronics box*
   - Length: 4”
   - Framing Type: T-Slot
   - T-Slot Framing Component: Rail
   - Rail Profile: Single
   - Single Rail Profile Style: Four Slot
   - Rail Height: 1"
   - Rail Width: 1"
   - Rail Construction: Solid
   - T-Slot Width: 0.256"
   - Center Hole Diameter : 0.205"
   - Material: 6105 Aluminum

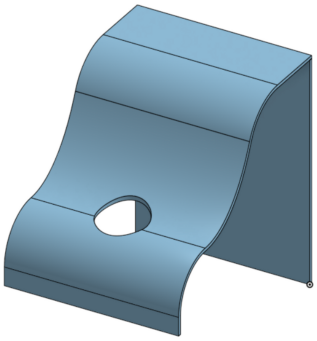
4. **Mounting L-Brackets (4x)** - *Connect side plates to top and bottom interior plates*
   - Material: PLA (50% infill)
   - Angle: 90°
   - Thickness: 0.25”
   - Leg Length: 1.44”
   - Width: 1.50”
   - Screw Clearance: 0.268”

5. **Mounting Corner Brackets (2x)** - *Connect top plate to back and side plates*
   - Material: PLA (50% infill)
   - Angle: 90°
   - Length: 3.622”
   - Width: 3.622”
   - Height: 0.875”
   - Screw Clearance: 0.320”

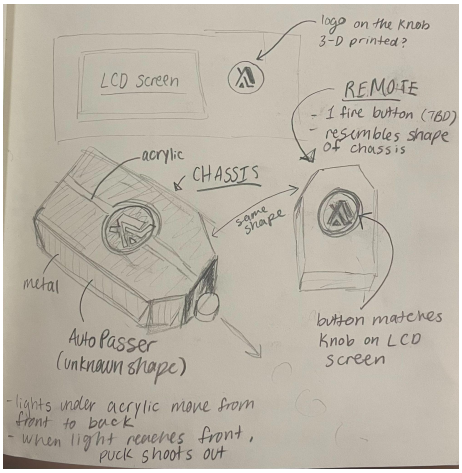
generate

Alternative Designs



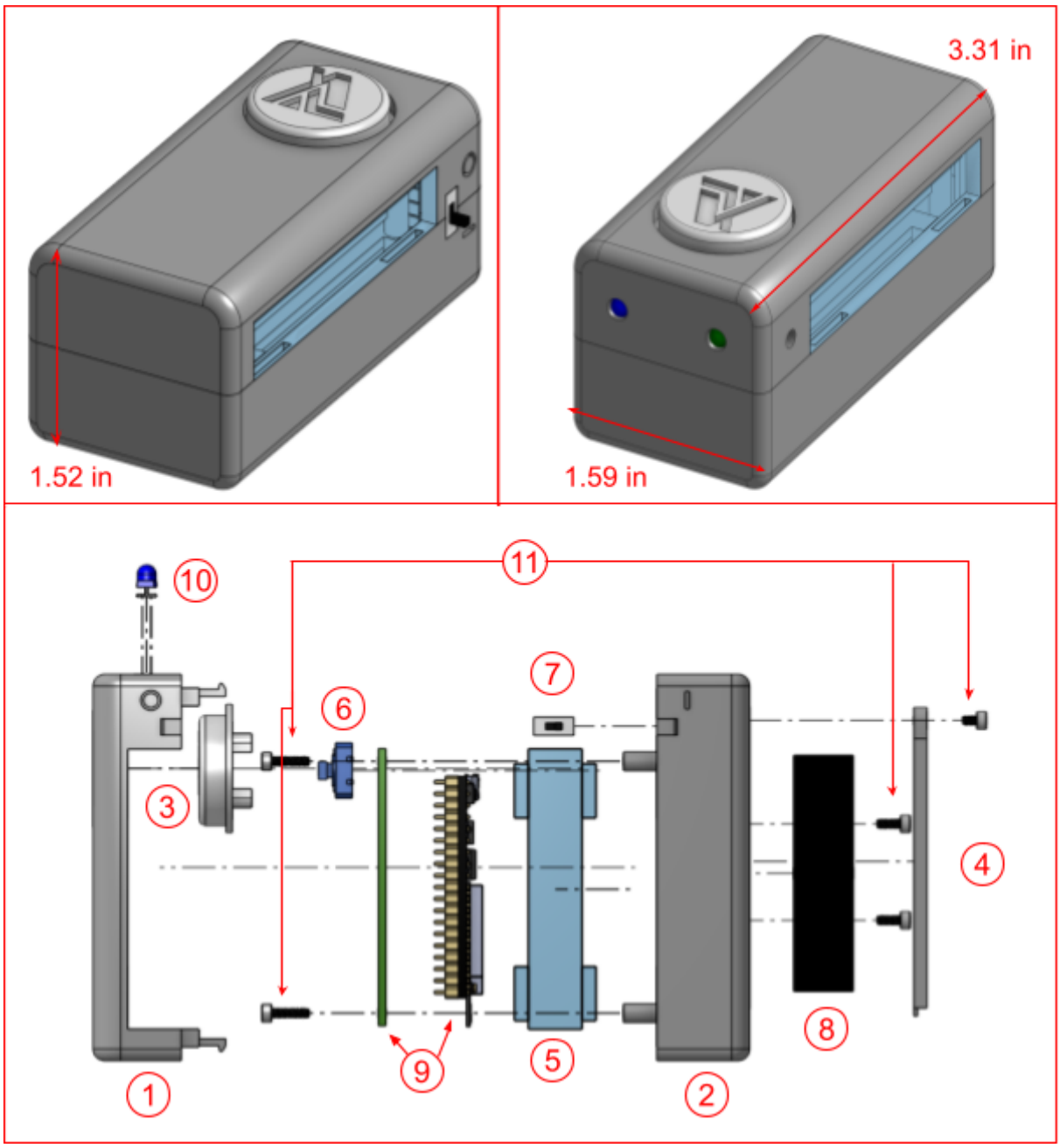


Another chassis design we considered featured a curved top. This design came out of brainstorming solutions to better incorporate the position of the loading column to look more intentional. We pivoted from this design as we determined acrylic to be the best material in our situation, and we did not have the tools to precisely bend the acrylic to the desired curves.

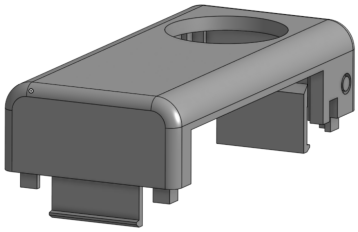


Another design element on the chassis that we explored were LED strips. Very early on in the chassis design, we had the idea of having LEDs along the chassis which would light up sequentially starting from the back to the front. When the light reached the front, a puck would shoot out. It provided the user a visual indication of when the pucks would fire giving the LEDs both a practical and aesthetic application. We were limited by the number of electrical pins and were ultimately unable to implement this.

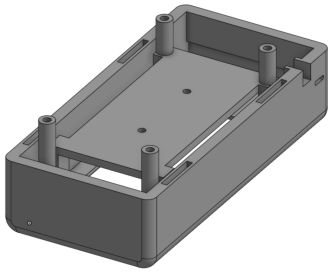
Remote Subsystem



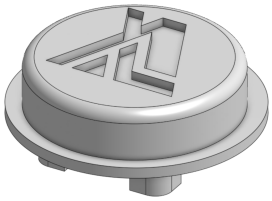
1. **Remote Control Top Chassis - Holds Button in place and includes snap-fit mechanism to hold remote together**
    - Material: Formlabs Tough 2000 Resin
    - Wall Thickness: 3mm
    - Maximum Length: 3.306 in
    - Maximum Width: 1.585 in
    - Depth (No snap-fit): 0.74 in
    - LED Through-hole Size: 4mm
    - Button Clearance Hole: 1 in
    - Snap-fit Dimensions (Calculated for Tough 2000 Resin):
        - H = 1.5mm, L = 8.25mm, y = 1mm



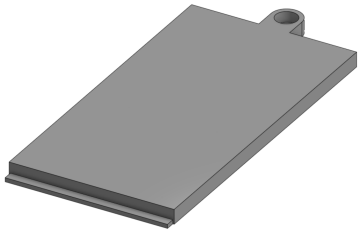
2. **Remote Control Bottom Chassis - Mounting for PCB, battery-pack, and bottom screw cover**
    - Material: Formlabs Tough 2000 Resin
    - Wall Thickness: 3mm
    - Maximum Length: 3.306 in
    - Maximum Width: 1.585 in
    - Depth (No stand-offs): 0.779 in
    - Through-holes Size: M2 (Tight Fit)



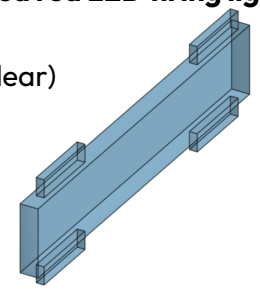
3. **Button Cover - Mounts to the button component that is soldered to the PCB for aesthetic and functional purposes**
    - Material: Formlabs Tough 2000 Resin (Change for aesthetic/feel preference)
    - Inner Diameter: 0.841 in
    - Outer Diameter: 0.941 in
    - Brim Diameter: 1.1 in
    - Stand-off height (To button component): 0.146 in
    - Tolerance in Button Hole (Top Chassis): 0.3mm
    - Logo Depth: 0.04 in
    - Manufacturing Note - May Require sanding of standoff for perfect fit



4. **Battery-pack Screw Cover - Seals in the battery-pack to prevent batteries from falling out secured by a screw on one side, and a lip on the other**
    - Material: Formlabs Tough 2000 Resin
    - Maximum Thickness: 2.75mm
    - Lip Thickness: 0.75mm
    - Lip Length: 1.7mm
    - Lip Clearance: 0.45mm (Side), 0.5mm (Bottom)
    - Counterbore Profile: M2 (Close fit)



5. **Clear Side-panels - Allow the user to see the internally mounted red LED firing lights whenever the fire command is sent**
    - Material: Formlabs High Temp Resin (Ideally Formlabs Clear)
    - Maximum Thickness: 3mm
    - Press-fit Tabs Thickness: 1.3mm
    - Press-fit Tolerance: 0.1mm
    - Maximum Length: 2.399 in



6. **Sparkfun Tactile Button Component - Detects user input**
    - Manufacturing Note - When soldering to PCB, make sure it is lined up as best as possible to the button cover hole in the top of the remote case

7. **Slider Switch - Turn off power going to PCB (Conserving Battery Life, etc.)**
    - Maximum Width:
    - Maximum Height:
    - Operating Life: 10,000 Cycles
    - Tolerance: ~0mm

8. **AAA Battery Pack - Holds two batteries that provides power to the PCB and other electronic components**
    - Batteries: AAA (x2)
    - Mounting Holes: M2 (x2)
    - Output Voltage: 1.5V

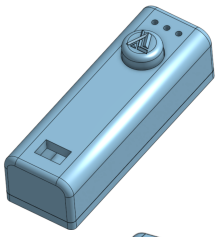
9. **Custom PCBA + ESP32 Microcontroller - Connecting all components, executing logic**
    - For details on the PCB design, see the PCBA Layout section of the report
10. **LEDs - To Indicate Status of Controller (Power, Connection, Firing Signal Sent)**
    - Color: 1 Green (Power), 1 Blue (Connection Status), 6 Red (Firing)
    - Type: SMD
    - Diameter: 3.9mm
    - Forward Voltage: 2.1V - 3.3V

11. **Mounting Hardware - Screws to hold PCBA, battery-pack, and back cover in place**
    - Size/Diameter: M2
    - Length: 3mm (x2) (Battery-pack), 5mm(x1)(Back cover), 8mm (x4)(PCBA)

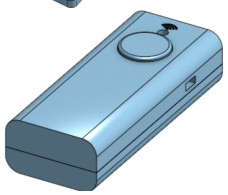
12. **ESP32 Antenna - Increases Range of Remote (Not in CAD)**
    - Manufacturing Note - There is a small through hole on the side to route the antenna cable into the casing, do this before engaging snapfits

Designing the Remote

Throughout the process, the design of the remote went through several iterations, each with tweaks and upgrades; this was mostly due to the vast amounts of small changes in components, as well as encountering limitations regarding what we could do. Below is a brief summary of each stage of the design process:

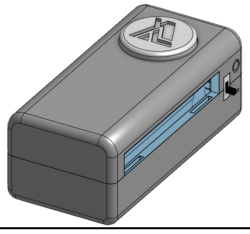


1. **All Hands Prototype -** Initially, we imagined the remote as a wii-mote sized device with a large,protruding button. This design ended up being extremely oversized for our components, as well as for a hockey drill. This did not feature snap-fits.

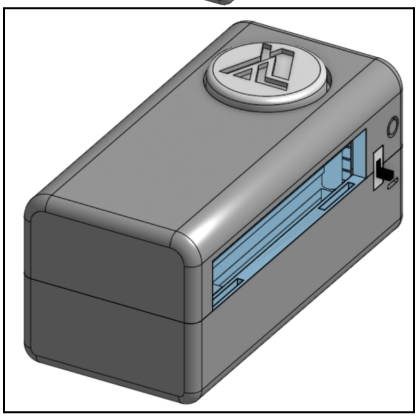


2. **Front Mounted LED Version -** First version of the controller to be fully rounded out due to the introduction of snap-fits. Originally a backlit button was supposed to be incorporated, leading to a longer remote. Room for our PCBA and a coin-style battery pack.

3. **Wide PCB Version -** Due to the completion of initial PCBA designs, the remote is noticeably wider. The design for our final version starts to become more apparent.



4. **Final Coin-style Version -** The smallest version in the whole design process for the remote; due to the tightening of tolerances and space inside the case, and the use of a coin-style battery pack, we had our smallest design. Also featured the addition of clear windows so the user can see the internally-mounted red firing LEDs.



5. **Final Design -** Due to the need for a higher voltage for our components, we had to scrap the coin-style battery-pack and switch to AAA. This caused a thicker overall design, which ended up being our final design

Material Selection and Manufacturing Recommendations

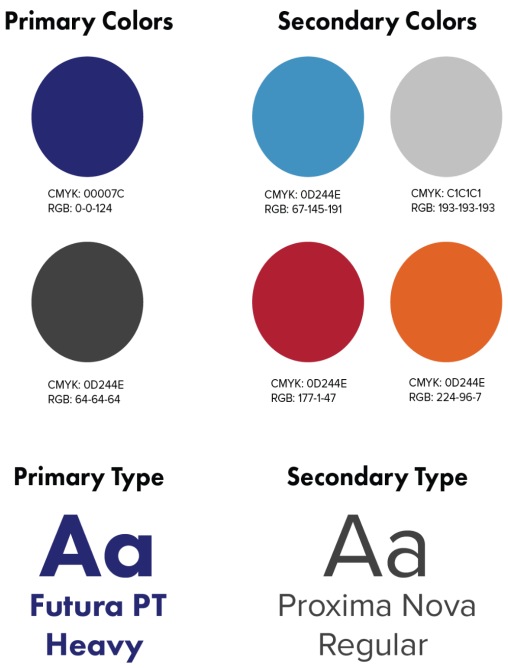
From the get go we knew we did not want to go with FDM printing for the remote because of the feel of layerlines. For that reason, on top of its durability and versatility, we chose to go with Formlabs Tough 2000 resin, especially since we were planning to incorporate snap-fits into the design.

For assembly, the only challenge was wiring the components properly and making sure the button and top chassis lines up; soldering has to be precise. Mechanically, the remote is very easy to assemble; the snap-fits go in easily and can be pried open with a screwdriver.

Branding

This semester, we expanded and iterated upon the branding for AutoPasser by designing logos, selecting a color palette, and choosing typography. We drew inspiration from the branding elements and incorporated them into our hardware and software designs.

AUTOPASSER



Color Palette

The color palette for branding was chosen very intentionally. Navy blue was chosen as a primary color as blue is very commonly associated with new technology. When searching for “high tech photo” in google images, almost every photo is blue [Fig 1].

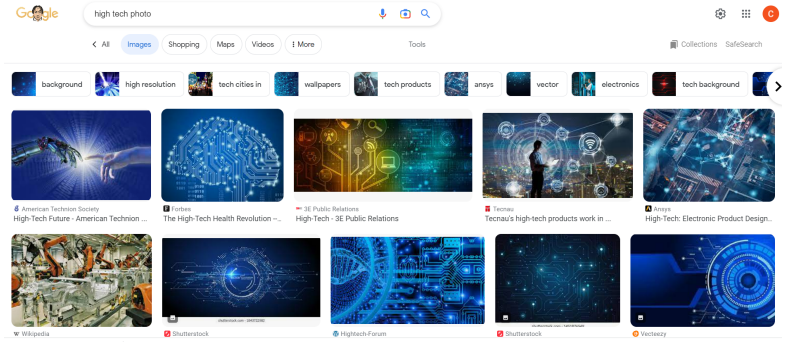


Fig 1: "high tech photo" Google Image search

Navy blue is also the second most commonly used color in NHL team logos [Fig 2].

Dark gray was chosen as a primary color. Most metals are gray, so we chose to include this color to be reflective of the machinery. This way the branding colors would be reflective of the actual product.

Light blue was chosen as a secondary color. It's common in design to have a lighter version of your primary color as your secondary color. Light gray was also chosen for this reason.

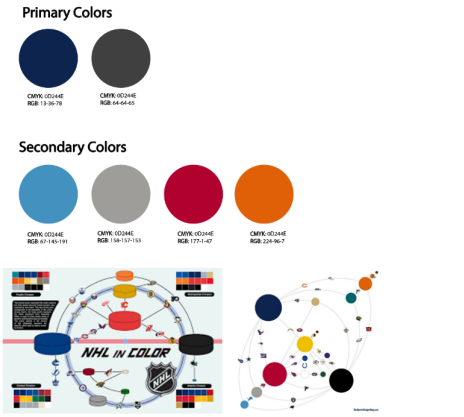


Fig 2: AP's colors next to common colors in the NHL

Red was the next secondary color chosen as it was the most used color in NHL team logos. The user for AutoPasser is someone interested in hockey so they are most likely familiar with NHL logos. We wanted AutoPasser's branding to fit this familiarity.

Orange was chosen as the last secondary color as it is a complementary color to blue [Fig 3]. In banding, complementary colors are often used for companies with a younger demographic. Complementary colors are often interpreted as fun and young. They are also commonly used in sports. This is seen in the branding for Fanta and Miami Dolphins [Fig 4].

Analogous colors, on the other hand, are commonly used for brands pushing to appear older and established. You can also see this with Mastercard and BP.

Fig 3: Complementary colors vs analogous colors on color wheel

Fig 4: Companies using complementary colors in branding

Type

The primary typeface selected for AutoPasser was Futura PT Heavy, a sans-serif font that has a contemporary feel. It is bold and structured which is aligned with the AutoPasser product. Proxima Nova Regular was chosen as the secondary type. It is also a san serif font that compliments Futura.

Logo Design

The logo combines the first letter “A” in AutoPasser, a hockey stick, and a puck [Fig 5]. The final logo was drawn with a 3-D feel for dimension and more of a “structured” look. Lots of sketching and iterations went into the logo design [Fig 6].

Fig 4: Final design for AutoPasser logo

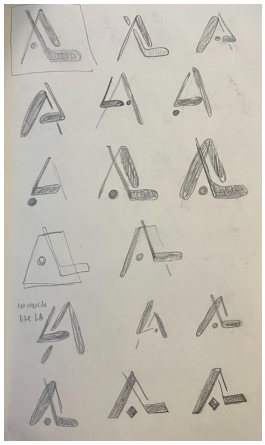
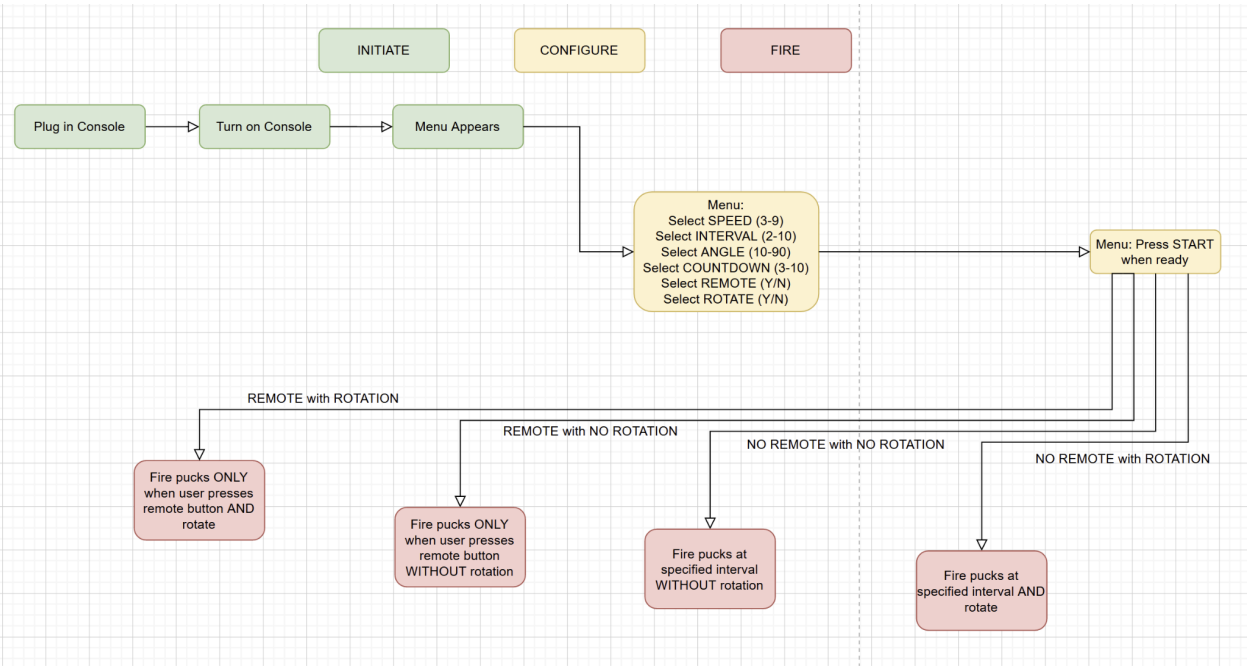


Fig 5: One page of early logo sketches

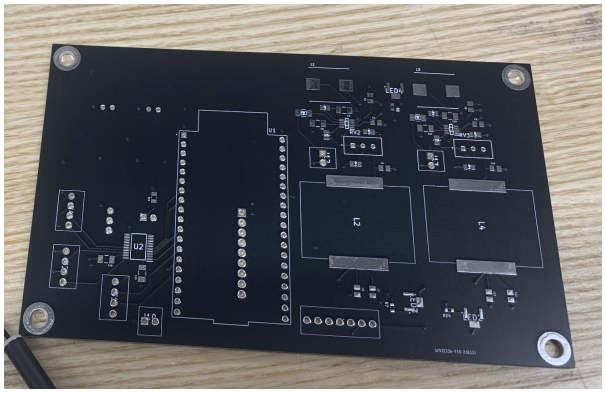
Electrical Design

Design Overview

The general design flow is shown below. This differentiates between the INITIATE, CONFIGURATION, and FIRING states.



Control System:



The control subsystem defined how every electrical component was tethered together in the circuit. This subsystem includes the console PCB and an additional Arduino Uno board, both of which communicate with each other via UART.

The initial design of this subsystem had everything incorporated in the PCB. However, the final prototype included the Arduino board to run the Neo motors and the screen due to incompatibility with the ESP32 microcontroller on the PCB. This explains some obsolescence in the PCB schematic and layout, and the specifics of the technical challenges that led to these design choices are detailed later in this report.

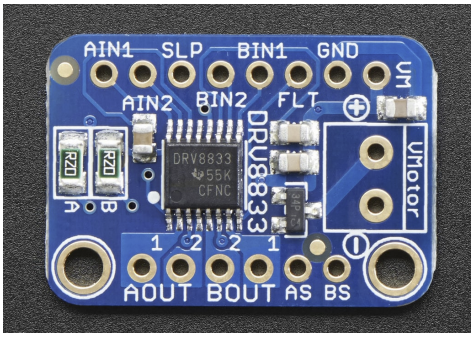
Firing System:

The firing system consists of 2 Neo Brushless Motors (V1.1) and 2 SPARK MAX Motor Controllers. These components are powered via a high-current 12V power supply and programmed via PWM signals from the Arduino Uno. The Arduino Servo.h library is used to generate the control PWM signals on specific GPIO pins.

The Neo motors are very current-hungry. They will work with any 12V power supply, but they can only run at high speeds and fire a puck if they are paired with a power supply that can provide a very high current. The power supply used in testing could source 100A; this is likely superfluous, but no current-limiting issues were encountered when testing the firing system, so it is a safe bet.

Turret System:

The turret system includes 1 4.5V Bipolar Stepper Motor (Pololu ID: 1475) and 1 Stepper Motor Driver (Adafruit ID: 3297). The stepper motor driver is controlled from the ESP32 in the console via 4 GPIO pins and the Stepper.h Arduino library. The driver is also sourced with the stepper motor voltage, in this case 4.5V. The stepper motor itself is connected to the driver via 4 wires. The 2 unconnected wires (yellow and white) are only used if the stepper motor is used in the unipolar mode.

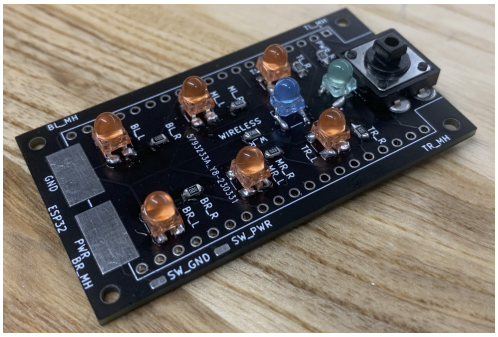


The stepper motor driver breakout board naturally inhibits the output current to around 1A. A resistor was soldered onto pads in the back to increase the current limit to 2A, which is the typical current per phase demanded by the stepper motor. Soldering an additional resistor onto the final bonding pads will remove the current limit entirely, but this is not recommended as it is simply not needed.

Remote System:

Just one remote interfaces with the AutoPasser console. This remote has one button, which is used to tell the console when to fire a puck. Red lights on the inside of the remote will flash to

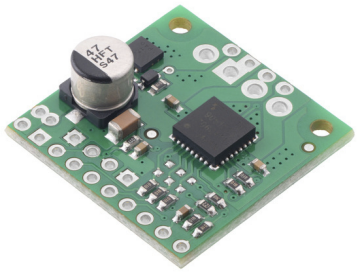
indicate that the firing signal has been registered. Inside the remote there is the ESP32 dev module, the remote PCB, and a battery pack containing 2 AAA batteries. The only hardware in the remote subsystem on the console side is an antenna attached to the ESP32 dev module; the rest is implemented through software.



The wireless protocol used in the remote subsystem is ESP-NOW. This is a WiFi communication protocol made specifically for the ESP32 architecture. Relevant test software is included in the Appendix.

Loading System:

The loading system has just 3 components: a linear actuator, an associated driver, and an ultrasonic sensor. The driver simply steps up an input signal to a 12V output signal. This is used to reverse the polarity of the 12V biasing on the linear actuator. The input signals are generated by the ESP32 in the console - they are simply DC signals generated from 2 pins switching HIGH and LOW. All useful documentation relating to the driver is included in the Appendix. The linear actuator itself is a very simple device - it just extends when provided with 12V and retracts when sourced with -12V.



The ultrasonic sensor is used to detect the presence of pucks in the loading column. This is used to determine when the console switches from the firing state back to the configuring state. Resistors were soldered with respect to the ECHO pin on this sensor to avoid sending a 5V signal back to the ESP32 and damaging the 3.3V device. The reasoning and specifics behind their placement is included in the Appendix.

User Interface:

The main method of interacting with the AutoPasser device is through the GUI on the main console box. The GUI is displayed on a 7-inch 40-pin TFT display (Adafruit ID: 2354), which is controlled by a separate driver board (RA8875). This driver board is then connected to the microcontroller in the main console. While we had planned to connect it to the on-PCBA ESP32, for an unknown reason (possible due to current limit), the display would not function properly, and thus we instead connected it to a Sparkfun Arduino Uno to control the UI.

The GUI layout and color scheme was based on FIGMA mock-ups by the team's designer, which was then replicated as accurately as possible through the Adafruit_RA8875 and Adafruit_GFX libraries.

The user interacts with the GUI via a button and potentiometer knob, which are placed to the right of the screen. The knob allows the user to scroll through the parameters displayed on the GUI, and the button allows the user to "click into" a parameter to be able to modify its value. After doing such, all other parameters will become grayed out and be unable to be modified, allowing the user to select the value of each parameter individually.

There are 4 scalable parameters displayed on the GUI (Interval, Speed, Angle, and Countdown). Interval (3-20s) represents the amount of time between individual pucks being fired, Speed (0-100%) represents the % power that the pucks are launched at, Angle (0-120 degrees) represents how much the firing angle will change between consecutive shots, and Countdown (3-20s) represents the amount of time from when the Start button is pressed to when the first puck will be fired. There are also 2 yes/no parameters on display: Remote and Rotate. Checking off Remote indicates that pucks will only be fired when the user presses the button on the remote. Checking Rotate indicates that the turret will rotate after every time a puck is fired.

Schematics and Layout:
All portions of the circuit schematic and PCBA layout process took place in KiCAD 6.0. Component libraries were downloaded from Mouser/Digikey/Ultra Librarian, and imported into KiCAD. Screenshots of the Schematics and PCBA Layout can be found below (Appendix: Schematics, PCBA Layout).

The remote PCBA was optimized for size, and contained bonding pads for the remote button, as well as various indicator LEDs to indicate Power on and Wireless connected status, as well as rows of leds to provide a brighter illumination for the Firing indicator. The PCBA also had bonding pads to which the battery was connected.

The main console PCBA included the two buck convertors (12V -> 5V, 4.5V) as well as a level shifter (3.3V -> 5V) and various screw terminals for connecting power and data wires. Indicator LEDs were also placed throughout this PCBA, with three on the front side (ON, Wireless Connected, and Firing), and 3 on the back (12V, 5V, 4.5V rails). Unfortunately, the 12V -> 4.5V buck convertor was not functioning as intended. This led to the introduction of an external buck converter module in the main console. Also, the IC for the level shifter was not functioning, which is why the Neo motors are controlled via the 5V-logic Arduino Uno and not the 3.3V-logic ESP32.

Name of Components

Most of the components used have already been mentioned in the previous sections, and this section will serve as a more in-depth description of each. As a general rule, all SMD components for PCBA were chosen to be Imperial size 0603+, with 0805 being preferred. If the PCBA were to be factory soldered, this would not be an issue.

1. NEO Brushless Motor V1.1: *Flywheel System*
    a. Nominal Voltage: 12V
    b. Empirical Stall Current: 105A
    c. Empirical Free Running Current: 1.8A
    d. Empirical Peak Output Power: 406W
    e. Motor Type: Brushless
    f. Required Component: SPARK MAX Motor Controller
    g. Datasheet

2. SPARK MAX Motor Controller: *Runs brushless motors*
    a. Nominal Voltage: 12V
    b. Operating Voltage Range: 5.5V - 24V
    c. Continuous Current: 60A
    d. Surge Current (2 seconds): 100A
    e. Signal Types: PWM, CAN, USB Control (through REV Hardware Client)
    f. Datasheet

3. LEDs: *Indicate voltage, wireless, firing*
    a. Links: Red LED, Blue LED, Green LED
    b. Forward Voltage: Red: 2.1V, Blue: 2.8V, Green: 2.8V
    c. Forward Current: Red: 20mA, Blue: 20mA, Green: 20mA
    d. Power Rating: Red: 120mW, Power Rating: 76mW, Green: 114mW
    e. Brightness: Red: 5.5cd, Blue 3.2cd, Green: 16.5cd
    f. Viewing Angle: Red: 30 degrees, Blue: 23 degrees, Blue: 33 degrees
    g. Datasheets: Red LED, Blue LED, Green LED

4. LED Resistors: *Limit current through LEDs, normalize brightness between different colors*
    a. Links: Red LED Resistor, Blue LED Resistor, Green LED Resistor (3.3V), Green LED Resistor (5V), Green LED Resistor (12V)
    b. Resistor Values: Red: 82Ω, Blue: 27Ω, Green (3.3V): 62Ω Green (5V): 287Ω, Green (12V): 1.2kΩ

   c. Power: Red: 0.125W, Blue: 0.125W, Green (3.3V): 0.125W, Green (5V): 0.2W, Green (12V): 0.2W
   d. Datasheet: Red LED Resistor, Blue LED Resistor, Green LED Resistor (3.3V), Green LED Resistor (5V), Green LED Resistor (12V)

5. ESP32: *Microcontroller for main console and remote*
   a. Operating Voltage: 5V
   b. Current Out Limit: 20mA per pin, ~0.75-8A total
   c. Wireless Protocol: WiFi 802.11, 2.4-2.5Ghz
   d. DataSheet

6. Sparkfun Redboard (Arduino Uno Replacement): *Controls screen and flywheel motors*
   a. Operating Voltage: 3.3V
   b. DataSheet

7. Level Shifter: *Increase voltage level of PWM signals (Did not end up working)*
   a. Input Voltage Range: A Pin: 1.2 - 3.6V, B Pin: 1.65 - 5.5V
   b. Intended Signal Voltage Shift: 3.3V -> 5V
   c. Enable pin must be tied to voltage high
   d. Datasheet

8. 7.0 Inch 40-pin TFT Display: *Display GUI*
   a. Resolution: 800x480
   b. Touch Enabled: Yes
   c. Dimensions: 165mm (W) x 100mm (H) x 4.9mm (D)
   d. Input Voltage Range: 3.0 - 3.6V, 3.3V ideal
   e. Connector Type: 40-pin FRC Cable
   f. Datasheet

9. RA8875 TFT Display Driver: *Control Display, interpret signals from microcontroller*
   a. Input Voltage: 3.3V
   b. Signal Type: PWM, SPI (Controlled through software)
   c. General Setup Guide
   d. Datasheet

10. Linear Actuator
    a. Operating Voltage: 12V
    b. Datasheet: Not available

11. Stepper Motor
    a. Operating Voltage: 4.5V
    b. Current Rating: 2A
    c. Steps per revolution: 200 (1.8 degrees per step)
    d. Datasheet

12. Stepper Motor Driver
    a. Max Continuous Current: 1.2A
    b. Max Peak Current: 2A
    c. Voltage Logic Range: 2.7V - 10.8V
    d. Datasheet

13. 100 Amp Circuit Breaker with Reset Switch: *Provide protection against excess current draw*
    a. Operating Voltage: 12-24V DC
    b. Current Limit: 100A
    c. Terminals: BAT for input wire, AUX for output wire
    d. Datasheet: Unavailable

14. 12 Gauge Wire: *Extension wires for motor cables*
    a. Current Limit: 40A
    b. Voltage Limit: 600V AC/DC

15. MP2315GJ-Z Step-Down IC: *Voltage step-down IC for buck convertor*
    a. Voltage Input Range: 4.5V - 24V
    b. Voltage Output Range: 0.8V - 22.8V
    c. Current Output: 3A
    d. Datasheet

16. Screw Terminals: *Intended to make it easier to connect/disconnect wires from PCBA*
    a. 2 Position Terminal, 3 Position Terminal, 4 Position Terminal, 7 Position Terminal, 9 Position Terminal
    b. Current Rating: 10A
    c. Voltage Rating: 150V
    d. Wire Gauge Range: 16 - 30 AWG

17. GUI Potentiometer: *Knob for controlling GUI*
    a. Power Rating: 0.5W
    b. Resistance Range: 10Ω - 5MΩ

   c. Voltage Limit: 600V
   d. Datasheet

18. GUI Button: *Intended to be button-knob combination, but potentiometer did not work*
   a. Datasheet

PCBA Components: *Based on Design here*

19. 100nF Capacitor
   a. Rated Voltage: 20V
   b. Datasheet

20. 22µF Capacitor
   a. Rated Voltage: 35V
   b. Datasheet

21. 4.7µH Inductor
   a. Rated Current: 9.5A
   b. Datasheet

22. 220µH Inductor
   a. Rated Current: 5A
   b. Datasheet

23. 20Ω Resistor
   a. Power Rating: 0.25W
   b. Max Voltage: 50V
   c. Datasheet

24. 7.5kΩ Resistor
   a. Max Power: 0.125W
   b. Max Voltage: 150V
   c. Max Current: 2A
   d. Datasheet

25. 20kΩ Resistor
   a. Max Power: 0.125W
   b. Max Voltage: 300V
   c. Datasheet

26. 75kΩ Resistor
    a. Max Power: 0.125W
    b. Max Voltage: 150V
    c. Max Current: 2A
    d. Datasheet

27. 180kΩ Resistor
    a. Max Power: 0.25W
    b. Max Voltage: 200V
    c. Max Current: 2A
    d. Datasheet

28. 200kΩ Potentiometer
    a. Resistance Range: 10Ω - 2MΩ
    b. Power Rating: 0.5W
    c. Datasheet

Alternative Designs

The majority of our design decisions were fairly straightforward, and we did not have to consider many alternative designs. Some decisions were carried over from the previous design (such as general wiring), and numerous others were heavily influenced by decisions made about the mechanical design of the system (what components were used to control the rotation and loading systems).

For the flywheel system, we cross-compared various different models of motors to determine which one offered the feature-set that we were looking for (namely, high power in a small form factor). Other models were mainly disqualified due to not providing as much power as the NEO V1.1s, or being overly large and bulky.

With regards to our PCBAs, for the most part their design was straightforward as well, and also hinged on decisions made for the mechanical design. Small changes such as the placement of individual components were made, but for the most part we were fairly confident in designing what was needed, and trying to minimize the number of unnecessary components.

Software Design

General Software Design

All code for the AutoPasser device was written in C++, through the Arduino IDE and Visual Studio Code. In order to upload code to the ESP32, a separate library (ESP32Servo, which can be searched for in the Arduino IDE) was required to be installed, and the device selected when uploading was the “DOIT ESP32 DEVKIT V1”. All code can be found in Appendixes below.

Explanation of Overall Software Decisions

GUI

The GUI was coded fully in C++, using the Adafruit_GFX and Adafruit_RA8875 libraries in Arduino. The software (Appendix: GUI Code) maintains an array of values representing all of the parameters (respective ranges for Interval, Speed, Angle, and Countdown, and a binary value 0 or 1 indicating whether the remote and rotation were on, and whether the start button had been clicked.

In the software, the display connection is initialized, and then the setupGUI() function is run, which creates the base layer of shapes and text. Additional functions are then run to update the display with the current parameter values, and after this is done, the code monitors input from the potentiometer and button to determine how to update. An outline is created around the currently selected parameter, and various helper functions were created to break the code down. Various precautions have also been implemented to ensure that each input is only registered once, and the correct values are being updated through the software.

All parameters have scaling that allows the potentiometer to be turned fully from one side to the other, and the on-screen bar is updated by redrawing the objects on the screen based on a percentage calculated from the parameter value. The Speed parameter has been scaled to operate in the 40-90% power range for the motors, as at least 30% power is required to fire a puck (40% used to ensure no jams), and capping the power at 90% greatly reduces the strain on the motors, and increases their efficiency, while providing minimal loss in speed of the puck. This scaling is done in software.

There are also three on-screen buttons, for enabling connection to the remote, allowing the device to rotate, and starting the machine. Unfortunately, we were only able to get the “Start” button to be functional, as while the remote and rotation buttons correctly update the status displayed, as well as the internal state of the machine, we were unable to complete their integration into the overall device software.

Considering that the GUI code is what is uploaded to the Arduino Uno, this code also includes the architecture for spinning the Neo motors. This occurs when the Arduino Uno sends the UI configuration data to the ESP32. The ability to *stop* spinning the motors when the ESP32 registers that there are no pucks left in the loading column has yet to be implemented. This involves the Arduino receiving a signal from the ESP32 via UART.

Console

The console-side code includes commands for every subsystem included in the AutoPasser console. This includes functions for stepper motor rotation, linear actuator extension/retraction, receiving wireless signals, detecting loaded pucks, flashing indicator LEDs. All software used throughout this project for the console is included in the Appendix. Everything is compiled as individual functions in the main code, but control for each component is broken up into individual scripts for easy code comprehension and debugging purposes.

Stepper motor control makes full use of the .step() method included in the Arduino Stepper.h library. This makes rotation very straightforward. Any mathematical computations performed with arguments for this method are simple conversions from degrees to steps and steps on the bigger turret gear to steps on the smaller gear.

Linear actuator control is achieved through simply alternating the HIGH/LOW status of 2 pins that are wired to the linear actuator driver.

Wireless signals are handled through the ESP-NOW communication protocol. The console antenna is only listening for signals when the user requests remote control and it is in the firing state.

Loaded puck detection is achieved through use of the ultrasonic sensor. Currently, the AutoPasser console continues to fire if there is something detected within 5 cm of the sensor and only fires once more if there is nothing detected.

Indicator LEDs are lit by pulling certain GPIO pins wired to LEDs HIGH. They are pulled LOW otherwise.

Remote

The remote-side code is the simplest and most straightforward of the software included in this prototype. Very simply, when the ESP32 controller detects a button push, it broadcasts a wireless signal that the console interprets as firing a puck. There is no danger of a button push triggering

the console when it is not suppose to; the console is programmed to listen only when Remote has been configured in the UI and it is in the firing state.

Communication

When looking at various modes of communicating within the machine, most of the decisions were straightforward. Specific methods of communicating between individual parts of the system can be found in the “Electrical Design” portion of the report.

As mentioned in that section, we chose to use the ESP-NOW protocol for wireless communication with the remote. ESP-NOW is a Wi-Fi based communication protocol developed by Espressif. The reason that this protocol was chosen is for a number of benefits when compared with other similar protocols. The benefits when compared to other similar protocols are:

Compared to:

- Bluetooth: ESP-NOW provided a more stable connection at longer distances, which was deemed more important than the power savings that could be offered by the BLE protocol.
- IR: IR is a directional protocol, meaning that properly receiving signals from any direction around the machine would require the user of multiple sensors, adding unnecessary complexity. Additionally, the line-of-sight requirement may provide minor annoyances depending on if there are any objects between the player and the machine.
- RF: The ESP-NOW protocol provided an easier implementation and higher capacity for data transfer, which while not a limiting factor at the moment, may prove to be so in the future.

Next Steps

Mechanical Design

- The loading column spring mechanism is not functional right now. The first step would be to reprint the top part of the column and get the mechanism working with the right springs.
- One issue that isn't quite accounted for is that the twisting motion to open the torsional spring also provides torque on the gear and will move the turret system slightly. The user can combat this by applying downwards force, but a redesign on the loading mechanism might be better. A mechanism that doesn't require torque to open up the column might be better. One of the other options was a camera lens type of mechanism. If this could be designed without the risk of rotating the turret then this would be a better option.

Electrical Design

- Unfortunately our power supply broke the week of Showcase. Currently, the machine is fully operational if an external power supply is used. The external power supplies in the makerspace aren't strong enough to spin the firing motors at an appropriate speed. Replacing the power supply should be the first step taken to get the machine up and running.
- The machine currently uses both an arduino and an ESP-32 on the PCBA. A worthwhile next step would be trying to get all the components to run off the ESP-32. If we had more time in the semester, our 2nd version of the PCBA would have addressed these issues.

Software Design

- Because the power supply broke before fully testing the machine, there may be some software debugging necessary once the machine is fully powered. Components were able to work individually, but there may be tweaks necessary to get everything working together.

Overall Project

- Truly the only main shortcoming is not having a working power supply. Once this is replaced, the machine should be fully functional, although some debugging EE/software wise may be necessary. Once this is in place, the client should build upon this machine however they want. The client mentioned implementing visual recognition software to track a player, which this machine should be capable of doing once the necessary components and software are in place.

Appendix

Bill of Materials (BOM)

[Full BOM can be found in this spreadsheet]

OTS parts:

MECHANICAL OTS PARTS				
PART NO	DESCRIPTION	SUB ASM	ORDERED?	LINK
OTS-M-001	4in. Hex Compliant Wheels (35, 50, 60 A)	Firing System	Using, ...	https://www.an
OTS-M-002	10-32 1" 50x Socket Screw	Firing System	Using, ...	https://www.m
OTS-M-003	10-32 100x hex nut	Firing System	Using, ...	https://www.m
OTS-M-004	1/2" Hex Shaft Collar	Firing System	Not Us...	https://www.ve
OTS-M-005	1/2" 3" Hex Shaft	Firing System	Using, ...	https://www.re
OTS-M-006	1/2" Flanged Hex Bearing	Firing System	Using, ...	https://www.an
OTS-M-007	1/2" Hex to Keyed Coupler	Firing System	Using, ...	https://www.an
OTS-M-008	Lubricated Turntable, Round	Turret System	Using, ...	1544T2
OTS-M-009	Tapered Heat-Set Inserts 10-32	Turret System	Not Us...	93365A154
OTS-M-010	10-32 Low-Profile SHCS	Turret System	Using, ...	93615A113
OTS-M-011	1/4-20 Low-Profile SHCS	Turret System	Using, ...	93615A410
OTS-M-012	1/4-20 kep nuts	Turret System	Using, ...	90675A029
OTS-M-013	Acrylic cement	Turret System	Not Us...	https://www.an
OTS-M-014	Needle-Roller Thrust Bearing - 1/4" Shaft Diameter, 11/16" OD	Firing System	Using, ...	5909K23
OTS-M-015	Needle-Roller Thrust Bearing - 5/8" Shaft Diameter, 1-1/8" OD	Firing System	Using, ...	5909k32

generate

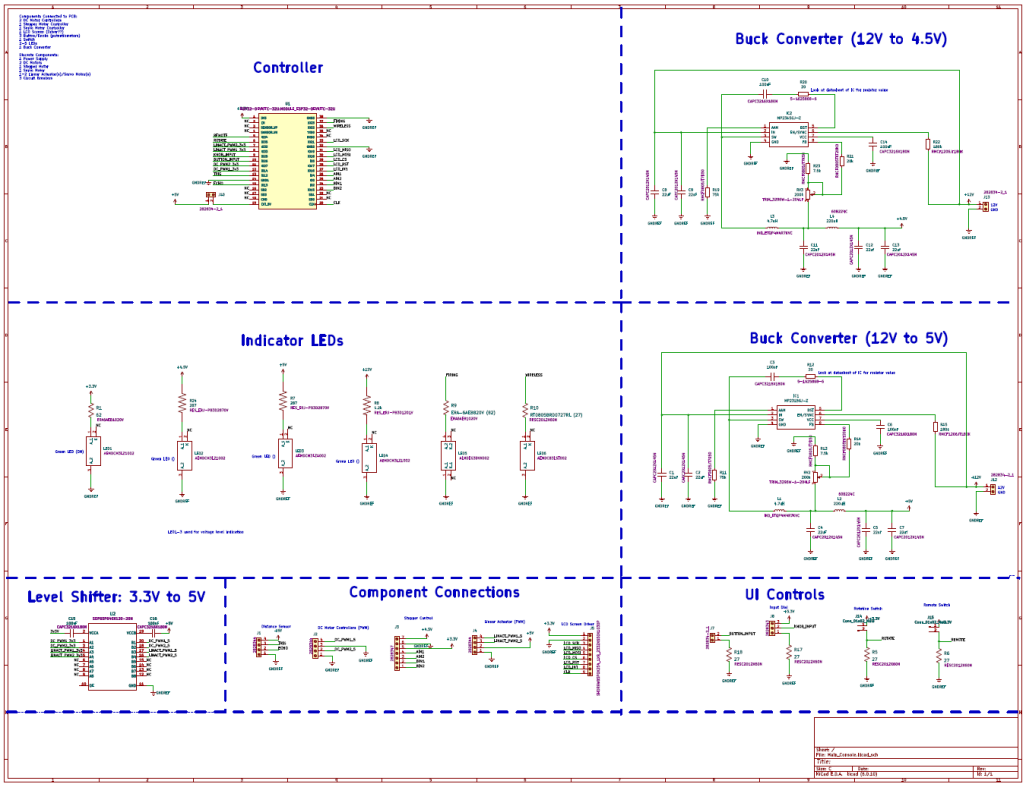
PART NO	DESCRIPTION	SUB ASM	ORDERED?	LINK
OTS-E-001	NEO v1.1 Motor	Firing System	Using, ...	https://www.re
OTS-E-002	SPARK MAX Motor Controller	Firing System	Using, ...	https://www.re
OTS-E-003	Green LED - 3.3V	Controls Box	Using, ...	https://www.di
OTS-E-004	Green LED Resistor - 5V	Controls Box	Using, ...	https://www.di
OTS-E-005	Green LED Resistor - 12V	Controls Box	Using, ...	https://www.di
OTS-E-006	Green LED Resistor - 3.3V	Controls Box/R	Using, ...	https://www.di
OTS-E-007	Blue LED	Controls Box/R	Using, ...	https://www.m
OTS-E-008	Resistor for Blue LED	Controls Box/R	Using, ...	https://www.di
OTS-E-009	Red LED	Controls Box/R	Using, ...	https://www.m
OTS-E-010	Resistor for Red LED	Controls Box/R	Using, ...	https://www.di
OTS-E-011	Flip Switch	Controls Box/R	Using, ...	https://www.m
OTS-E-012	ESP-32	Controls Box/R	Using, ...	https://www.m
OTS-E-013	Level Shifter	Power Delivery	Using, ...	https://www.di
OTS-E-014	LCD Screen Driver Board	Controls Box	Using, ...	https://www.ac
OTS-E-015	LCD Screen	Controls Box	Using, ...	https://www.ac
OTS-E-016	Dial Potentiometer	Controls Box	Using, ...	https://www.ac
OTS-E-017	Switched Circuit Breaker	Power Delivery	Using, ...	https://www.an
OTS-E-018	10 feet of 12 AWG Wire	Power Delivery	Using, ...	https://www.m
OTS-E-019	22uF 35V Capacitor	Power Delivery	Using, ...	https://www.di
OTS-E-020	100nF 50V Capacitor	Power Delivery	Using, ...	https://www.m
OTS-E-021	DC-DC Step-Down Module	Power Delivery	Using, ...	https://www.di
OTS-E-022	4.7uH 9.5A Inductor	Power Delivery	Using, ...	https://www.di
OTS-E-023	220uH 5A Inductor	Power Delivery	Using, ...	https://www.di
OTS-E-024	75K Resistor	Power Delivery	Using, ...	https://www.di
OTS-E-025	200K Potentiometer	Power Delivery	Using, ...	https://www.di
OTS-E-026	7.5K Resistor	Power Delivery	Using, ...	https://www.di
OTS-E-027	20K Resistor	Power Delivery	Using, ...	https://www.di
OTS-E-028	Linear Actuator	Loading Systen	Using, ...	https://www.an

Fabricated parts:

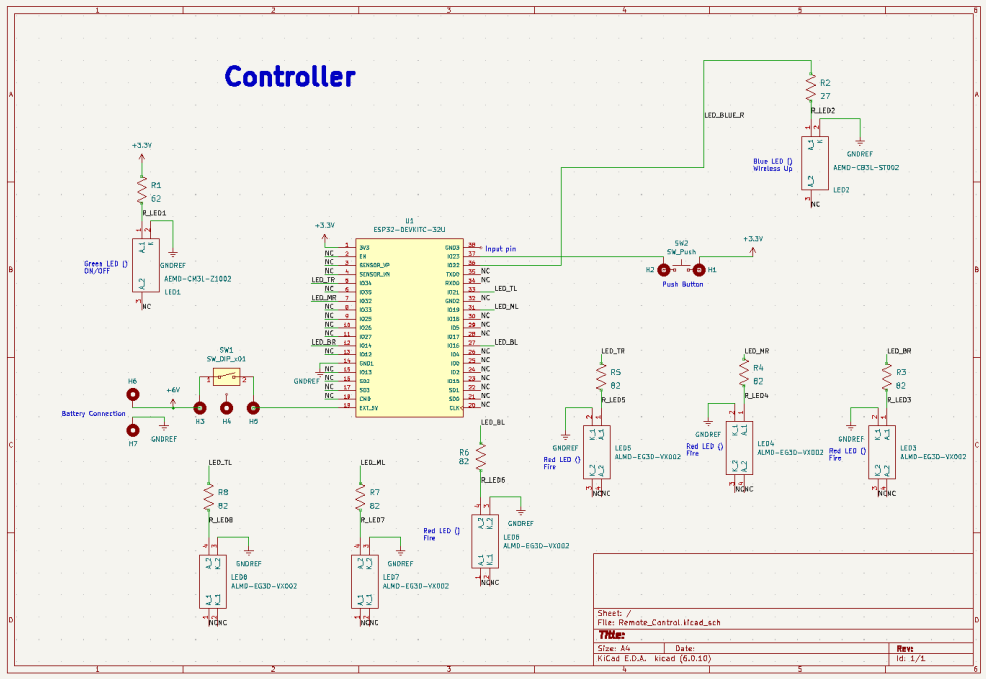
PART NO	PART NAME	DESCRIPTION	ADDITIONAL INFO	MFG PROCESS	VENDOR
FAB-001	CIRCULAR-MOTOR-MOUNT	Supports the motors & aligns shafts with bottom plate		PLA 3D Print	N/A
FAB-002	CIRCULAR-BRIDGE	Joins the motors mounts in to constrain them & provide stability		PLA 3D Print	N/A
FAB-004	REMOTE-TOP	top half of remote		SLA 3D Print	N/A
FAB-005	REMOTE-BOTTOM	bottom half of remote		SLA 3D Print	N/A
FAB-006	REMOTE-BUTTON	Button for remote		SLA 3D Print	N/A
FAB-007	REMOTE-BATTERY-PACK	battery pack cover for remote		SLA 3D Print	N/A
FAB-008	CHASSIS-PLATES	Top, back, and 2x side plates fo		LASER CUT	N/A
FAB-009	BASE-PLATE-1	Botton acrylic base plate, secur		LASER CUT	N/A
FAB-010	BASE-PLATE-2	Top acrylic base plate		LASER CUT	N/A
FAB-011	LOADING-COLUMN-1	Bottom part of loading column,		PLA 3D Print	N/A
FAB-012	LOADING-COLUMN-2	Top lip of loading column used		SLA 3D Print	N/A
FAB-013	LOADING-COLUMN-3	Top part of loading column witl		PLA 3D Print	N/A
FAB-014	DELRIN-GEAR	Delrin gear for turret system		CNC	Xometry

Schematics

Main Console:

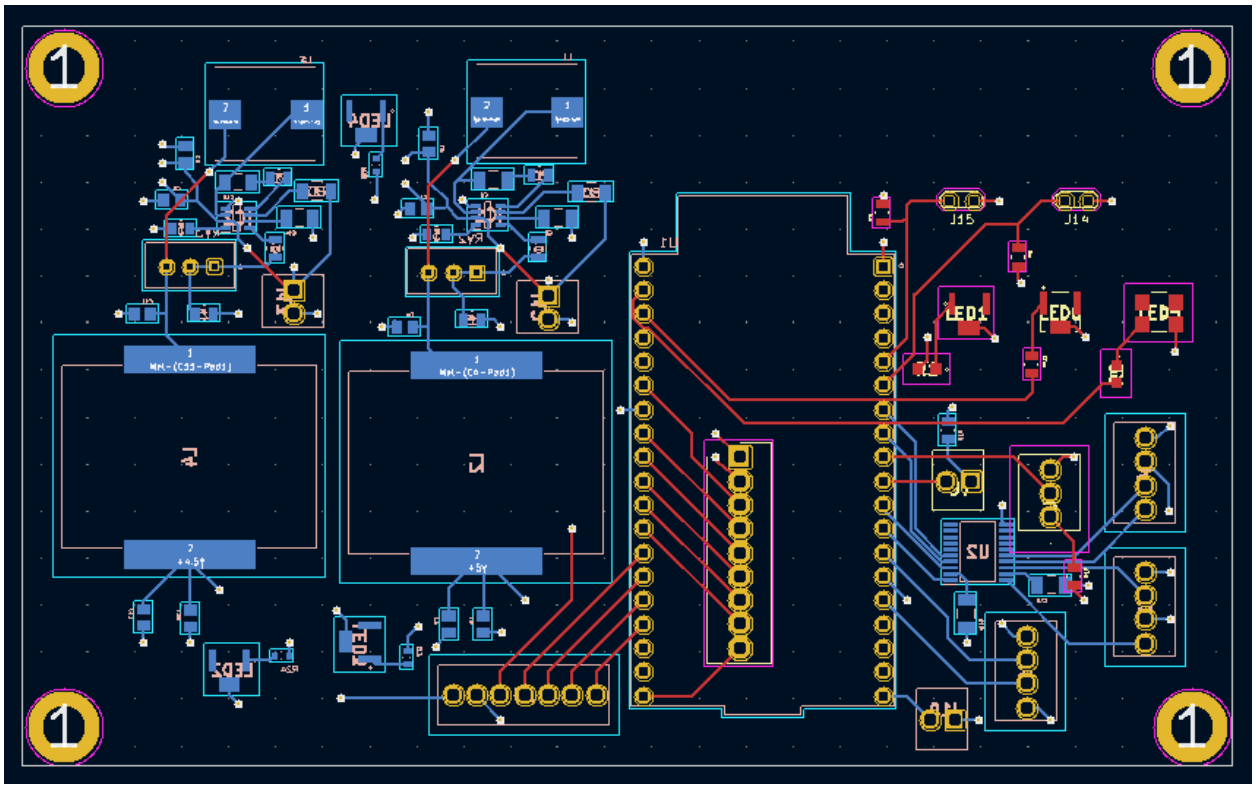


Remote Control:

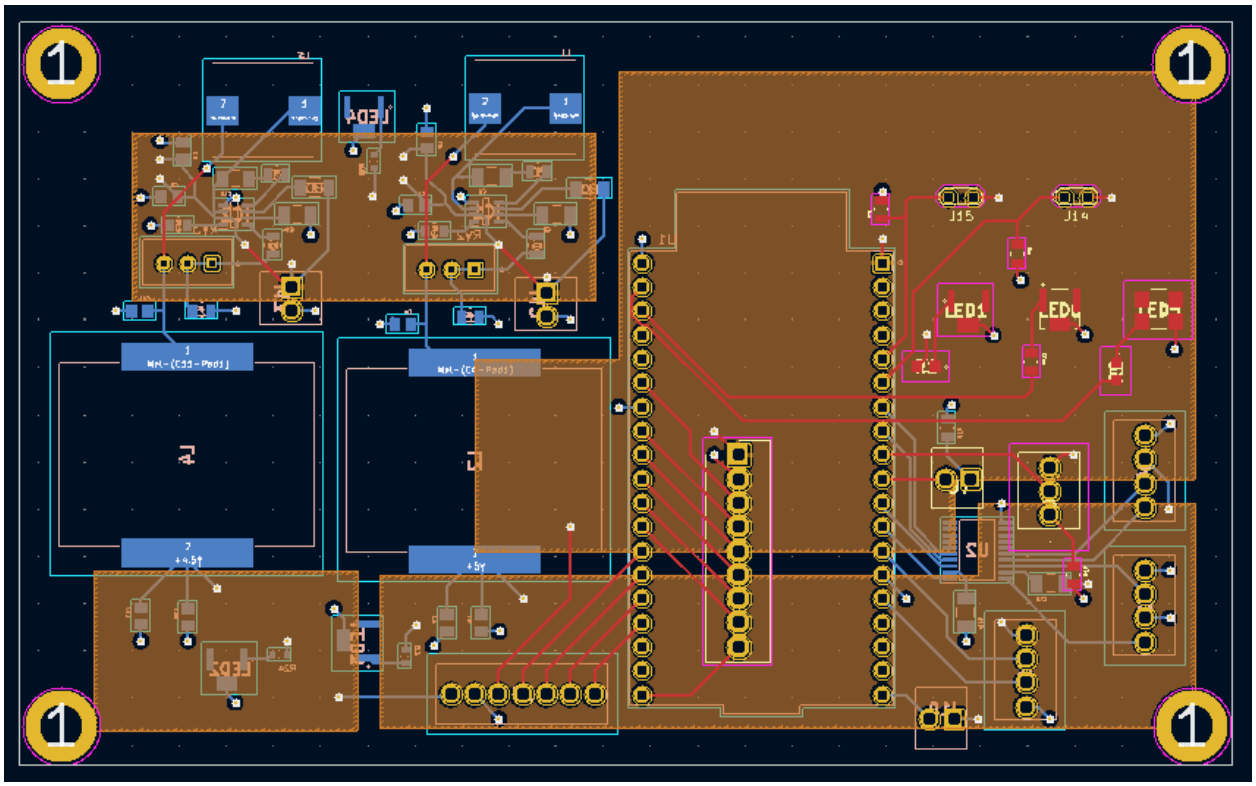


PCBA Layouts

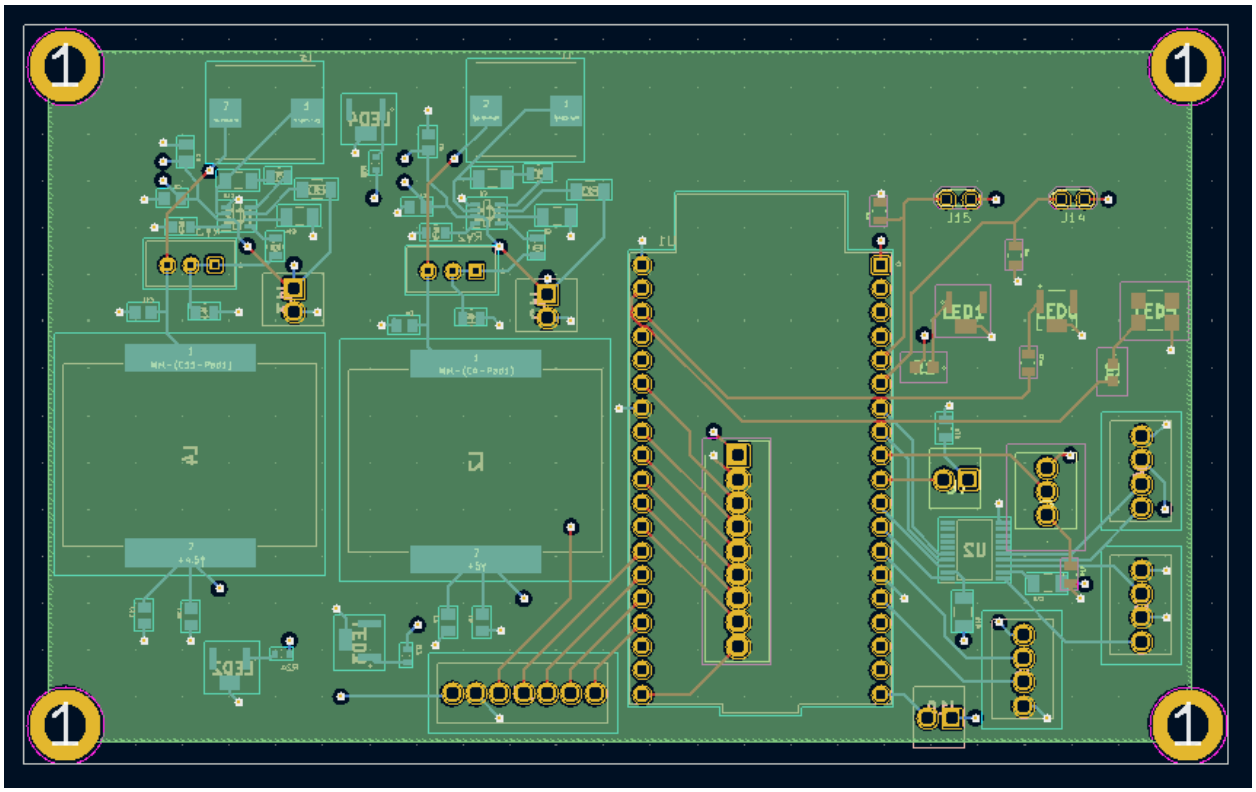
Main Console (Power/Ground planes hidden):



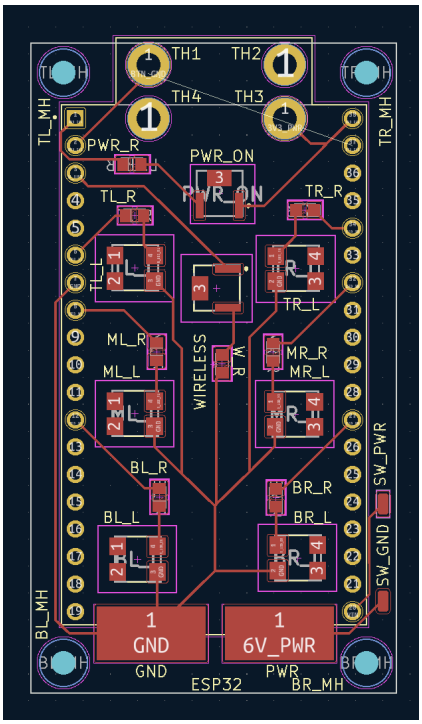
Main Console (Power plane visible):



Main Console (Ground plan visible):



Remote Control:



General Code

GUI Code

GUI.ino was used for majority of time, however stopped working after implementing additional functions. In order to be able to reference those functions later, working code was transplanted into displaytest.ino.

Main - GUI.ino: Link
Testing - displaytest.ino: Link

Console Code

Main - Functional_Main_Recv_ScreenData.ino: Link
Stepper Motor - Adafruit_stepper_motor_code.ino: Link
Proximity Sensor - Test_ProxSensor.ino: Link
Control Neo Motors - Working_with_Seria.inol: Link
ESP32 UART - Test_UART_ESP32Side.ino: Link
Arduino UART - Test_UART_ArduinoSide.ino: Link

Control Linear Actuator - new_linear_actuator.ino: Link

Remote Code

Main - Remote_Main.ino: Link
ESP-NOW Send Side - send_fire.ino: Link
ESP-NOW Receive Side - receive_fire.ino: Link

Hardware Documentation

ESP32 Dev Module: Link
Linear Actuator Driver: Link
Linear Actuator: Link
Stepper Motor Driver: Link
Stepper Motor: Link
Neo Motor Controller: Link
Neo Motor: Link
Ultrasonic Sensor 3.3V Modification: Link